

The Effects of Increasing US Clean Electricity Subsidies on Electricity, Emissions, and Welfare

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Abstract

Utility-scale clean electricity subsidies have long been the centerpiece of US federal climate policy, yet little is known about how changing subsidy levels affects electricity capacity, emissions, and welfare. I estimate these marginal effects by exploiting the spatial variation from a recent place-based subsidy, the Energy Community Bonus. The subsidy causes large increases in solar and battery capacity, with smaller gains in wind capacity. Long-run effects, which account for generator location choice, institutional hurdles in the several-year process of building a new generator, and inter-project externalities, are even larger than the short-run effects. These physical effects combined with large wedges between pre-existing subsidies and social marginal costs implies a marginal increase to clean electricity subsidies drives significant welfare gains. The Energy Community Bonus amplifies the marginal welfare and emissions effects of solar and wind subsidies, but dampens the effects of battery subsidies.

In 2022, the Inflation Reduction Act (IRA) expanded subsidies for utility-scale clean electricity to be worth an estimated \$400 billion through 2035 (Bistline et al., 2023b). In 2025, the One Big Beautiful Bill Act (OBBBA) surprisingly repealed the subsidies for solar and wind electricity, which account for about half of future planned clean capacity, keeping the subsidies in force for the other half, mostly batteries. Despite the enormity of these policies and the policy changes, little is known about the physical and welfare effects of clean electricity subsidies. This is because there has historically been no quasi-experimental variation in subsidy levels.¹ Meanwhile, pundits speculate that nowadays the subsidies are largely or entirely inframarginal, citing permitting delays, local opposition to nearby generators, and institutional barriers (NYT, 2023; WSJ). Renewable electricity companies and interest groups claim they do not need the tax credits to be profitable, while policymakers insist the subsidies will be re-enacted anyway (Politico, 2026b). Nonetheless, engineering models like the one that produced the \$400 billion estimate suggest clean electricity subsidies have large marginal effects and are the single most impactful US federal climate policy (Bistline et al., 2023a).

Part of the IRA reform, the “Energy Community Bonus,” made clean electricity subsidies place-based. The Energy Community Bonus is intended to steer clean electricity to Energy Communities, areas with local economies reliant on fossil fuels. I use this place-based subsidy variation to estimate subsidy effects on clean electricity capacity and welfare. This same place-based subsidy that provides the variation for this exercise also changes the answer to the question of marginal welfare effects. Traditionally, combining distributional policy objectives with externality-correcting efficiency mechanisms necessarily entails an efficiency-equity trade-off (Okun, 1975). But in this case, it is not obvious whether the place-based subsidy improves or worsens efficiency. This is because the pre-existing uniform subsidy is an imperfect externality price. Incorporating the place-based incentive could, in principle, move clean electricity subsidies closer to or farther from an efficient price. Combining my empirical estimates with geographic and electricity emissions data, I calculate the marginal welfare effects of clean electricity subsidies and assess whether the place-based subsidy improves or worsens its welfare effects.

My empirics focus on the window of time after the Energy Community Bonus was enacted in

¹Even a recent cutting-edge meta-analysis of US climate policies, Hahn et al. (2026), offers no insight on the marginal effects of utility-scale solar subsidies because, “...there are no quasi-experimental estimates of the impact of utility-scale solar [subsidies].” There are also no estimates of battery subsidy effects, given it is an emerging technology. For wind subsidies, the meta-analysis uses the best available estimates, which are based on data covering 1998 to 2007, but wind turbine technology, markets, and electricity institutions have changed substantially since then.

2023, and before the surprise repeal for solar and wind was announced in summer 2025. I estimate the short-run effects of the policy on new renewable generator construction during that time. However, the process of building a new clean electricity generator takes several years. So, the developers responsible for that new clean capacity likely chose their generator locations years before the Energy Community policy was even conceived. In order to calculate long-run policy effects, I exploit the institutional requirement that all proposed projects must enter an “interconnection queue” to connect to the national electricity grid. The value of estimating the long-run effect estimate is that, without the long-run analysis, it is not clear the long- and short-run effects have the same sign, let alone magnitude. The interconnection queue process is rife with inter-projects externalities, both positive and negative, that could limit or bolster the subsidy effects on clean electricity capacity.

I find large marginal effects of the place-based subsidy on new capacity additions, despite the fact that new generators did not endogenously choose their location according to the subsidy. Solar and battery capacity additions drive the large result. There is no detectable effect on wind electricity capacity for this short horizon. Over two years of interconnection queue data turns out to be sufficient to infer subsidy effects on electricity capacity that will materialize—or would have materialized were the subsidy not repealed—years in the future. The long-run results accounting for endogenous location choice and the interconnection queue externalities are even larger. The place-based subsidy more than triples capacity additions in Energy Communities in the long run.

Estimates in hand, I use the marginal value of public funds (MVPF) to measure the marginal welfare effects of marginal clean electricity subsidies. I use the “marginal” qualifier because I do not estimate the effects of the base value of the subsidies, I only estimate the effect of an incremental change to the base value. The MVPF is a particular benefit-cost ratio, societal willingness to pay (WTP) per net fiscal dollar expended. Across all technologies, I find societal WTP for a marginal fiscal dollar toward clean electricity is \$6.49. The MVPF for solar is nearly 9, for wind is 7, and for batteries is 3. The lower battery value is due to emissions from charging, but the value is still quite high because the electricity consumed when they charge tends to be cleaner than the electricity displaced when they discharge. To some extent, the large empirical estimates drive the result. Even more so, the large MVPFs are due to the large wedge between the externality value and subsidy levels.

The Energy Community Bonus raises the MVPF for solar and wind. This is not because Energy Communities tend to be sunnier or windier. In fact, the opposite is true. Instead, the Bonus raises the MVPF because, on average, a marginal generator in an Energy Community displaces dirtier electricity than a marginal generator elsewhere. This mechanism does not imply the same unambiguous efficiency improvement for batteries. It turns out that a battery in an Energy Community is, on average, more

emissions intensive than a battery elsewhere, so the place-based subsidy lowers the subsidies’ marginal societal value.

I use a sufficient statistics framework to calculate the welfare effects of the Energy Community Bonus relative to a capacity-neutral policy, one that results in an equal amount of electricity capacity as the place-based policy, but uniformly distributes subsidy across the country. The Energy Community Bonus for solar and wind increases welfare by \$800 million in present-value every year it is disbursed. However, the annual welfare loss for batteries is even larger, a \$1.1 billion loss per year.

One concern for using spatial variation to measure subsidy effects, and underscored by the large estimates, is the possibility of spillover effects. I pay close attention to this, describing the physical and economic mechanisms by which subsidizing clean electricity in Energy Communities might increase or reduce clean electricity capacity elsewhere. I use a simple theory model to show why it is implausible that economic spillovers are large in magnitude, despite the large effect on capacity additions. This does not rule out infrastructure-based spillover effects, though. Comparing Energy Communities to neighboring areas, I find that, if anything, the Energy Community Bonus *crowded-in* clean electricity in nearby areas that do not receive the Bonus. The effect is not present in the long-run estimates, and is excluded from the welfare calculations. Nonetheless, it suggests that the large results could be conservative.

Given the fame of the Energy Community policy and the importance of the surrounding questions, there are already other early studies on the Energy Community Bonus, and surely more to come. Balkir (2026) and Keuzenkamp et al. (2026) also estimate the Energy Community Bonus effects on new operating projects during the early years of the policy, also finding a positive result, in addition to estimating employment and political election effects. Zhang (2025) employs a calibrated economic model for a forward-looking perspective on Energy Community Bonus effects. Doshi et al. (2026) estimate the Bonus effects on new proposed projects in the interconnection queue, finding the Energy Community Bonus has a null or negative effect on queue entry, which contrasts with the result here. They also use a structural model of generator location choice to model subsidy effects. I use transparent empirical strategies—both simple difference-in-difference (DID), as well as a synthetic DID (SDID)—to estimate physical subsidy effects in the short-run. Like Zhang, I also do a forward-looking looking analysis, and like Doshi et al., I use empirics. I measure how projects fare after entering the queue. This analysis involves multiple sources of theoretical ambiguity and proves to be quantitatively important. Lastly, I differ in extending my empirical estimates to measure the emissions and welfare effects of this place-based policy and clean electricity subsidies generally.

I also contribute to the empirical research on clean electricity subsidies in the US. Previous work

compares wind turbines receiving output subsidies and investment subsidies, showing that output subsidies cause the generators to generate more electricity (Aldy et al., 2023). I incorporate their finding in my model of emissions abated by clean electricity receiving output subsidies. Hitaj (2013) and Metcalf (2010) are early studies testing whether US clean electricity subsidies cause more wind capacity. I revisit wind capacity effects of these papers, exploiting quasi-experimental variation, and contribute estimates for solar and battery storage. Hahn et al. (2026) uses these estimates of wind elasticities to generate estimates of the MVPF for wind. I extend their results to solar and batteries, update the results for wind using contemporary elasticities, and investigate how the place-based policy affects welfare.

Finally, I contribute to the research on efficiency-equity and emissions-equity trade-offs. In the case of carbon pricing, it is clear that incorporating distributional mechanisms worsens efficiency (Goulder et al., 2019). In the case of dirty energy, it is clear that a subsidy entails a welfare loss (Hahn and Metcalfe, 2021). In those contexts, the crucial question is at what welfare cost. Conversely, in the case of subsidies for clean goods, that welfare cost can be positive or negative because these subsidies are almost never priced proportionately to the externalities the clean goods generate.² Linn (2022) shows that means-testing a subsidy for electric vehicles improves the subsidy cost-effectiveness because lower income households are less likely to be inframarginal buyers. While it is true that in both that instance and the present one the distributional mechanism improves efficiency on net, every instance of this is ambiguous and an empirical question.

I next describe the Energy Community policy and the process of building clean electricity generators in more detail. I pay special attention to policy details that could threaten identification, such as Energy Community designation or anticipating the 2025 policy repeal. I also provide greater detail on the interconnection queue process as it pertains to what long-run effects are measurable using the relatively short two-and-a-half year time window. Section 2 describes the data, Section 3 presents the empirical strategy and results, Section 4 the welfare and emissions model and results, and Section 5 concludes.

²By clean subsidies, I refer to subsidies for substitutes for emissions-producing products. Of course, a cost-effective carbon abatement subsidy is uniform. I also do not refer to a second-best optimum, where it is not clear that the efficient clean subsidy is positive (Fullerton and Stechuk, 2026). Most clean good subsidies are well within the Pareto frontier.

1 Background

1.1 Clean Electricity Subsidies and the Energy Community Bonus

The US federal government has long offered two nationwide tax credits for clean electricity. The Investment Tax Credit (ITC), enacted with the Energy Tax Act of 1978, subsidizes construction costs. The Production Tax Credit (PTC), placed in force by the Energy Policy Act of 1992, subsidizes 10 years’ worth of electricity generation on a per kilowatt-hour (kWh) basis. The credits cannot be combined, so electricity project developers must choose which to claim.

The IRA set the ITC to 30 percent of construction costs, and the PTC to 0.3 cents per kWh in 1992 value, adjusted to inflation (0.6 cents per kWh in 2026) (CRS, 2026).³ For 2023 and 2024, only certain clean electricity technologies were eligible for each tax credit.⁴ Beginning in 2025, due to the Inflation Reduction Act, the tax credits became “technology neutral,” such that any clean electricity is eligible to claim the credits regardless of technology.

The Energy Community Bonus is a larger increase in the subsidy value for the ITC than the PTC. It increases the ITC by 10 percentage points (30 to 40 percent) and the PTC by 10 percent (0.6 to 0.66 cents per kWh in 2026). Table 1 lists the three types of definitions of Energy Communities.⁵ An area only needs to meet one of these definitions to be an Energy Community, so the full set of areas receiving the subsidy is the union, not the intersection, of these definitions.

Figure 1 shows the locations of Energy Communities in the US. The first definition in the table and the top-left map are areas with a sufficiently high historic fossil fuel employment rate and a high previous-year unemployment rate. The areas designated ECs by this definition can and do change year-to-year, as local unemployment rates and the national unemployment rates change. Many counties have this designation for at least one year, especially in the southwest, Texas, and the Rust Belt. Fewer counties have the designation in all three years (shown in dark red, the combination of the semi-transparent colors). Changes in the local and national unemployment rates drive the substantial year-to-year changes in designations. A project retains permanent Energy Community status if it

³These values assume the project meets wage and apprenticeship requirements for 10 years. They exclude the value of the possible domestic content bonus, which is worth the same as the Energy Community Bonus.

⁴Technologies that are eligible for both the ITC and PTC include solar, wind, municipal solid waste, and geothermal electric. Only eligible for the ITC are energy storage, fuel cells, combined heat and power, microturbines, microgrid controllers, and interconnection property. Only eligible for the PTC are hydroelectric, biomass, and landfill gas.

⁵According to the IRA law text, there is another type of Energy Community not listed in this table: areas with sufficiently high local tax revenue coming from fossil fuel industries. However, Treasury has never identified the data defining local tax revenues, so the policy is not in effect and its definition is moot.

<i>Category</i>	<i>Spatial unit</i>	<i>Criteria</i>
Fossil fuel employment and unemployment	Groups of counties	Fossil fuel industry employment higher than 0.17 percent at any time after December 31, 2009 and a local unemployment rate higher than the national unemployment rate in the previous year.
Coal closure	Census tracts (subdivisions of counties)	Contains a coal mine closed after December 31, 1999 or a coal-fired electricity generating unit retired after December 31, 2009, or a tract spatially neighboring such a tract.
Brownfields	Parcels (subdivisions of census tracts)	Brownfield sites, based on a definition specific to the Energy Community Bonus.

Table 1: Energy Community definitions.

begins construction while its location is designated as an Energy Community. As I explain in more detail below, a project usually chooses its location several years before beginning construction. In empirical analyses measuring the subsidy effect, I treat only areas that meet this definition every year as Energy Communities, since the areas experiencing year-to-year variation may experience the persistent subsidy effect.

The other two Energy Community definitions—coal census tracts and brownfield parcels—are permanent designations, so that particular concern does not apply. Coal Energy Communities are areas that have had a historic coal mine or coal-fired electricity closure. They are scattered throughout the eastern half of the US, and cover large swaths of Appalachia and the Mountain West. One concern with this definition is that a coal-fired electricity generator closure could perhaps be endogenous with renewable electricity development. Perhaps the closure causes newly uncongested transmission lines or substations that make these areas good places to build renewables. Or, perhaps a closure signals a location is a particularly unprofitable one for electricity generation. I run robustness checks dropping this type of Energy Community in the empirical analyses, and the results are unchanged. Coal mine closures do not carry the same issue because they close due to reasons unrelated to renewable suitability, such as increasing mine costs and national trends in natural gas and electricity prices (Watson et al., 2023).

The final definition of an Energy Community is a brownfield. While the Brownfields map in Figure 1 has much more colored area than the other two maps, the Brownfields underlying the colored areas are small, as opposed to the other two maps, where the entire colored areas are Energy Communities.

According to the IRA law, the Energy Community Bonus, along with all clean electricity subsidies, was set to phase out in 2034 or when the US cut emissions 75 percent below 2005 levels, whichever

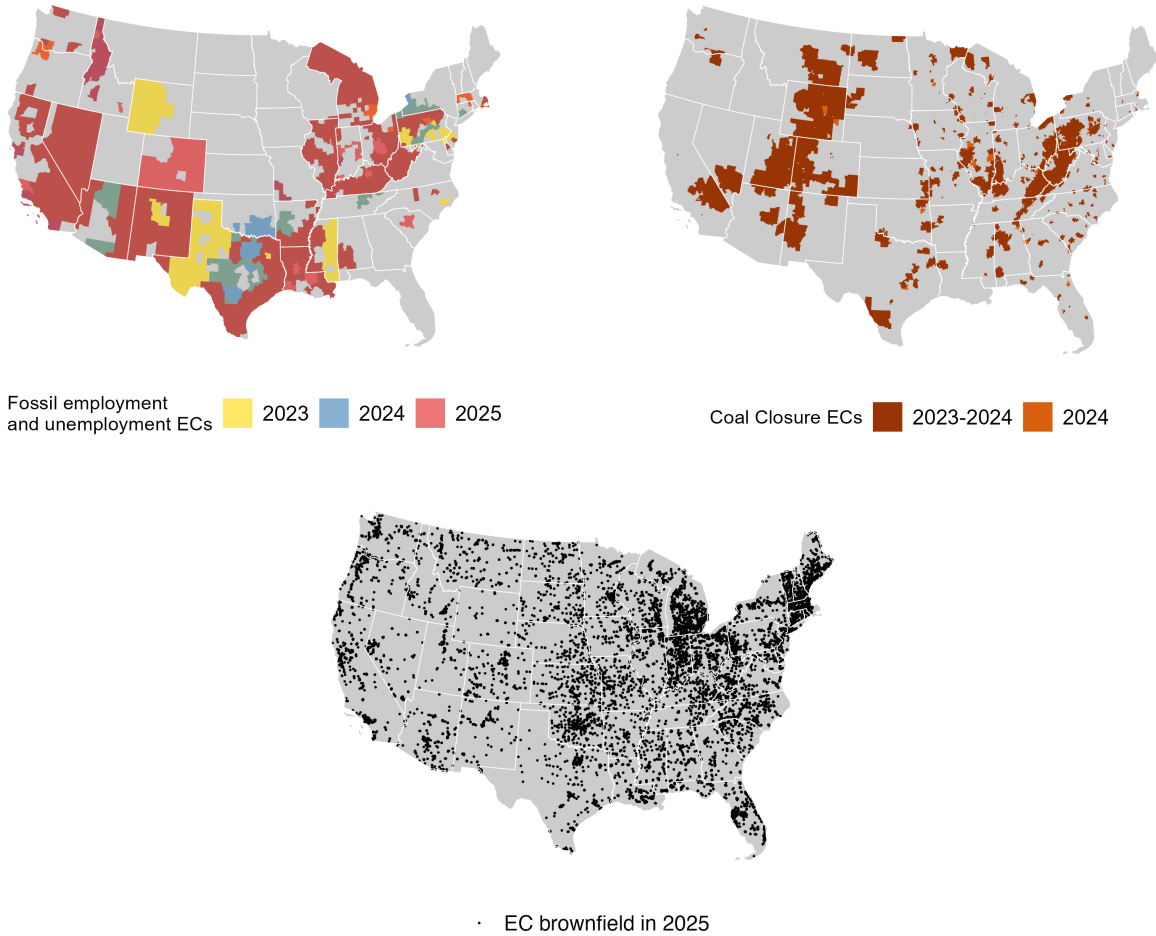


Figure 1: Energy Community (EC) locations. Brownfields are shown as point locations and are not to scale.

happens first. However, the OBBBA accelerated that timeline for solar and wind technologies. I next describe the repeal in more detail.

1.1.1 Partial Subsidy Repeal

The OBBBA, announced July 2025, two and a half years after the Energy Community Bonus enactment, accelerates the phase-out of the ITC, PTC, and Energy Community Bonus for solar and wind technologies. Solar and wind projects now must begin construction before July 4, 2026 to be eligible for the tax credits.⁶ On the other hand, the ITC and PTC phaseouts are unchanged for battery storage, hydroelectricity, geothermal, nuclear, and all other clean technologies. Furthermore,

⁶A project is also eligible if construction begins after July 4, 2026 and is placed in service by December 31, 2027, but that is widely considered an unrealistically fast turnaround from beginning construction to operation.

to be clear, these phase-out deadlines only apply to new projects. For example, a project that claims the PTC before the policy lapses is guaranteed the full 10 years of subsidy, even after the policy lapses for new projects.

The analyses in this paper focus on the two and a half years between the Energy Community Bonus enactment in 2023 and policy repeal in 2025. The remainder of this section considers whether repeal anticipation by project developers could affect the estimates of the Energy Community Bonus effect.

With respect to “beginning construction” before July 2026 for tax credit eligibility, for most projects this means satisfying a “physical work” test, such as pouring foundation or buying custom components.⁷ Industry groups estimate over 200 GW of solar capacity and 30 GW of wind capacity have satisfied this requirement as of mid-2026. This is about all of the solar and wind capacity additions anticipated through 2030, even according to optimistic estimates that take the IRA tax credits for granted (Mackenzie, a,b; SEIA). Therefore, regardless of whether project developers anticipated the repeal, the repeal is unlikely to affect any new generators coming online through the late 2020s, let alone those in the short-run capacity addition analyses using data from 2023 through 2025.

Repeal anticipation could, in principle, affect the long-run estimates of the Bonus effect on project planning, though. In particular, estimated subsidy effects could be biased down, as developers undervalue the subsidies understanding they may not be available for their project during the relevant time, when they begin construction. To some extent, the large results in this paper undermine this concern.

Furthermore, qualitative evidence from late 2024 and early 2025 suggests the repeal of the ITC and PTC for solar and wind was widely considered unlikely.⁸ An often-cited justification for the Inflation Reduction Act’s subsidies over other possible climate and energy policies was its perceived political durability. Repealing the Inflation Reduction Act provisions would require an act of legislation, which would effectively require a Republican trifecta. One outlet predicted a joint probability of a Republican trifecta of about 23 percent on election night.⁹ Even after the November 2024 election, when a Republican trifecta materialized, it was still highly uncertain what would come of the ITC, PTC, and Energy Community Bonus. Many Republican congressmembers favored the clean electricity

⁷Before OBBBA, the laxer “safe harbor” rule allowed projects to begin construction by spending 5 percent of anticipated construction costs. However, OBBBA requires the more stringent physical work test for projects that claim to begin construction between September 2025 and July 2026.

⁸Unfortunately, for quantitative evidence, I am not aware of any prediction markets where bettors and industry experts could speculate on the repeal of clean electricity tax credits.

⁹50 percent chance of Republicans winning the presidency, 49 percent the House, and 92 percent the Senate, according to 538 on November 5th, and assuming independence of the events (538).

subsidies (Politico, 2026a). In January 2025, experts speculated that the clean electricity subsidies were among the least likely Inflation Reduction Act subsidies to be repealed, in part because many of the subsidies were for projects in Republican-leaning areas (Fujii-Rajani and Patnaik, 2025).

The importance of repeal anticipation on project planning also depends on how costly it is for project developers to plan and change plans. If it is inexpensive to plan and change plans, then repeal risk is small, even if it materializes. In that case, developers’ decisions would be little affected by repeal anticipation. The next section offers more insight here, describing the basic process of planning and building new clean electricity generators in the US, and the costs incurred at each stage of planning. As it turns out, the early stages of planning are low cost, but planning costs grow in later stages.

1.2 Building a Clean Electricity Generator in the US

Roughly, building a new clean electricity generator happens in three stages over several years. First, the project is planned. The project developer chooses a location, technology type, and project size. Second, the project enters an “interconnection queue.” An interconnection queue is a process by which projects learn how much it will cost to connect to the national electricity grid. The cost is highly uncertain before this process, and most projects withdraw from the queue upon learning their costs. Finally, should the project developer choose to accept its costs, it finishes acquiring the necessary permits and financing, constructs the generators and transmission lines, and is placed in service.

My first empirical analysis is simply on the final outcome: the MW of capacity added to the grid. The rest of the empirical analyses measure how the subsidy affects interconnection queues. This has the advantage of capturing how project developers value the Energy Community Bonus while endogenously accounting for the subsidy value in their decision to locate in an Energy Community. However, interpreting the data and the estimates requires some institutional knowledge. The rest of this section describes the relevant details to define my outcome variables and interpret the results.

Interconnection queues are managed by local grid operators, each with their own set of rules and management systems, in addition to overarching rules from the Federal Energy Regulatory Commission (FERC). One common feature across the various interconnection queues is that it is not especially costly to enter an interconnection queue, at least relative to other generator development costs and expected future revenue upon being placed in service. Entry costs are measured in tens of thousands of dollars, while interconnection costs are routinely tens of millions.¹⁰

¹⁰To enter the queue, a developer just needs to demonstrate land ownership or an option to purchase land, and/or pay a mostly-refundable deposit. The rules vary by local grid operator. For example, California Independent System

My first outcome variable for estimating the long-run subsidy effects is interconnection queue entry. More projects entering interconnection queues in Energy Communities is a precursor for more future generators in Energy Communities. However, most projects that enter the queue ultimately withdraw. So this effect does not alone represent the long-run effect of a subsidy on future generators.

While a project is on the interconnection queue, it receives a series of increasingly accurate interconnection cost estimates, based on “interconnection cost studies” conducted by the grid operator. After each cost study, a project has a fixed time window, usually about a month, to accept or reject the costs. If the costs are higher than the project is willing to accept, it withdraws from the queue. Grid operators are regulated entities with no profit incentive. The costs are simply the result of engineering estimates of the infrastructure costs required to accommodate the new project, such as new and upgraded transmission lines and substations.

A project’s interconnection cost is highly uncertain before it undergoes this series of studies because a project’s infrastructure requirements and resultant costs depend on what other projects enter the queue, and how those projects react to their estimated costs. If another project withdraws from the queue, that could lower costs for other projects due to requiring less ambitious grid upgrades, or it could raise costs for other projects, as grid upgrade costs are split among fewer projects.

Given these inter-project withdrawal externalities, grid operators impose penalties for withdrawal, which increase as the project advances through the queue process. In late stages, the penalties can reach millions of dollars. However, large study-to-study cost estimate changes are common. To limit project developers’ risk of exposure to extreme cost estimate changes, grid operators often include caveats such as no withdrawal penalty if interconnection costs more than double.¹¹

The last two outcome variables in my empirical analyses are the accepted interconnection cost levels and the probability of accepting an interconnection cost study. Interconnection costs are one

Operator (CAISO) allows entrants to pay a \$500 thousand to \$2 million refundable deposit instead of demonstrating land control. (CAISO, 2024). Midwest Independent System Operator (MISO) charges a \$7,000 application fee, plus \$8,000 per MW for the interconnection study.(MISO, 2026). PJM requires generators to pay a deposit ranging from \$10 thousand to over \$300 thousand depending on the generator size, but only 10 percent of this deposit is not refunded if a project is withdrawn early in the queue process. Otherwise, the deposit is returned less 10 percent and the study costs, which are tens of thousands of dollars.(PJM, 2026)

¹¹For example, for Independent System Operator of New England (ISO-NE), after the first study the penalty is two times the cost to conduct the study (tens of thousands of dollars), 5 percent of estimated network upgrade costs after subsequent studies (the “cluster re-study” phase), 10 percent of estimated network upgrade costs after a late-stage study (the “facility studies” phase), and 20 percent of estimated network upgrade costs if a project withdraws after it signs an interconnection agreement. If the cost estimate increases over 25 percent in an early study, or if the estimate more than doubles after a later study, then there is no withdrawal penalty (ISO-NE, 2025).

mechanism by which the subsidy may affect future generation, if projects have a higher willingness to accept due to the subsidy. However, the inter-dependence of projects' interconnection costs could also be a mechanism that limits the subsidy effect on electricity capacity. If more capacity additions mean more expensive infrastructure upgrades, the supply curve of new electricity generators could be convexly increasing.

The probability of acceptance measure captures the net effect of willingness to accept and inter-project externalities. It is also theoretically ambiguous, even without inter-project externalities. If the subsidy causes more queue entry, the marginal entrants may be lower-value projects than the inframarginal entrants and tend to reject their costs more often.

After a project completes its cost studies, it signs an “interconnection agreement.” Not enough time elapsed between policy enactment and repeal announcement to measure the Energy Community Bonus effect on outcomes after the cost studies, such as interconnection agreement signings or construction, in a way that endogenously accounts for location choice. These milestones are usually three or more years into a project's lifecycle. Instead, to quantify outcomes for inframarginal projects that did not choose their location considering the subsidy value, but still may be affected by the subsidy after their location decision, I simply look at capacity additions as the outcome variable. However, given the increasing withdrawal penalties, most withdrawal happens early in the queue process. Therefore, I quantify long-run Energy Community Bonus effects without observing interconnection agreements and later stages, as this captures most of the possible subsidy effect.

The next section describes the data I use to assess how the subsidy affects new capacity additions, interconnection queue entry, interconnection costs, and interconnection withdrawal probability. After describing the data, I turn to the empirical strategy and results.

2 Data

This section describes the data sources, with more detail on data cleaning in Appendix [A](#). After that, and before the empirics, I specify here how I define Energy Communities, which addresses a data hurdle caused by a mismatch between the spatial resolutions of the interconnection queue data and Energy Community definitions.

2.1 Data Sources

New Electricity Generators

New capacity addition data are from Energy Information Administration (EIA) 860M forms. The data include the generator’s technology type, capacity in MW, and point location. My sample begins in 2017 and ends in December 2025.

Interconnection Queue Entry

I obtain queue entry data from Interconnection.FYI.¹² The data include project-level observations of queue entrants, including their technology type and MW capacity. Unlike the point-located EIA generators, most queued projects are only spatially precise to the county level. The queue data cover all seven major ISOs and RTOs, plus 49 non-ISO utilities, accounting for about 97 percent of US electricity generation (Rand et al., 2025). I aggregate the data to the county-level since this is the most precise spatial unit that covers all available areas. I drop some areas due to insufficient queue entry data, shown in Appendix Figures 9 and 10. Most notably, I drop the Electric Reliability Council of Texas (ERCOT), which covers most of Texas. My sample begins in 2017 and goes through June 2025, just before the repeal announcement.

Interconnection Cost Studies

Interconnection cost study data are available for Southwest Power Pool (SPP), MISO, and PJM. Other grid operators have some study data available, but no others have sufficient historic data from before and after 2023. Interconnection.FYI provides the SPP and MISO data. PJM data come from a combination of Interconnection.FYI, Lawrence Berkeley National Laboratory (LBNL), and my own web-scraping. For a sense of the scope of this coverage, Appendix Figure 11 shows each grid operator’s coverage. Fortunately, these ISOs cover many Energy Communities and non-Energy Communities. The cost data are at the study-level, recording the study phase, the date of the study, and the quoted interconnection cost. I merge this to the other queue dataset, which also includes information on dates that projects withdraw, to infer whether projects accept or reject their estimated interconnection costs.

¹²Interconnection.FYI provides the data for the popular Queued Up report from the Lawrence Berkeley National Laboratory (LBNL) (Rand et al., 2025).

Energy Communities

Fossil fuel employment thresholds are defined using US Census County Business Patterns (CBP) for 2010 to 2024 (Census, 2025). Local unemployment rates are calculated using the Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS) for years 2022 and 2023, to define Energy Communities in years 2023 and 2024, respectively (BLS). I identify census tracts with coal mine closures and coal generator closures using the official data from the US Department of Energy (DOE) for tax years 2023 and 2024 (DOE et al., 2023, 2024).

There is no publicly available data on all potential EC Brownfields that meet the Internal Revenue Service (IRS) EC Brownfield definition, but there is public data on EPA Brownfields from the EPA Facility Registry Service (EPA, 2025). I use these data as my Energy Community Brownfield set, dropping some observations that I identify as unlikely to qualify for the Energy Community Bonus based on the law text (more on this in the Appendix).

Covariates

I use various covariates to get a sense of how Energy Communities compare to the wider US and the donor pool of potential control counties. The 2021 Census American Community Survey provides population estimates and the 2022 Quarterly Census of Employment and Wages for average wages. Direct normal irradiance (DNI, or sunniness) is from the National Solar Radiation Database (NSRDB), summarized at the maximum pixel value in each county (Sengupta et al., 2018). Wind speeds at 100m from the WIND Toolkit are summarized in the same way (Draxl et al., 2015). Finally, transmission line data come from Our Grid Future, and 2020 land cover data from the National Land Cover Database (NLCD) via IPUMS NHGIS.

Data for Welfare and Emissions Analyses

To calculate emissions and welfare effects, I need a variety of data to model how a marginal unit of clean electricity affects emissions. Most of these data come from National Lab of the Rockies Cambium model (Gagnon et al., 2024). I also use solar and wind capacity factors from the ReEDS model and technology costs from the Annual Technology Baseline (Sergi et al., 2024; NLR, 2024). The next section details the empirical analysis and results, which do not rely on these data, so I defer an in-depth description until Section 4, after the empirics.

3 Subsidy Effects on Electricity Capacity

This section presents the estimation strategies and results for the subsidy effect on clean electricity capacity. The effects on capacity additions and queue entry are both measured in MW per county, so the model for these two estimation strategies is the same and the analyses are presented together. Analysis of projects' accepted interconnection cost levels and probability of accepting costs is at the project level instead of the county level, and presented after. First, I describe how I define Energy Communities and non-Energy Communities that can serve as controls. The analyses all use this definition.

3.1 Identifying Energy Communities and non-Energy Communities for the Empirical Analyses

The main specifications of my empirical analyses define Energy Communities as counties that meet the fossil fuel employment designation in all years since the policy was enacted or are completely covered by coal mine or coal generator Energy Communities. I consider non-Energy Community counties to be those that have never met the fossil fuel employment designation, have zero coal-eligible census tracts, and fewer than ten Brownfields. I require counties be fully covered by eligible Energy Community areas because the queue data are at the county level, while coal mine, coal generator, and Brownfield Energy Communities are all defined at sub-county levels (Table 1). In this way, I omit counties that have the most uncertainty over whether the projects they contain qualify for the Energy Community Bonus.

It is impractical to drop all counties with a brownfield from the controls, since this would leave very few potential control counties. The cutoff of ten is arbitrary, but the results are insensitive to more or less restrictive cutoffs. If anything, including Brownfields-qualifying projects in the control group would bias the results toward zero.

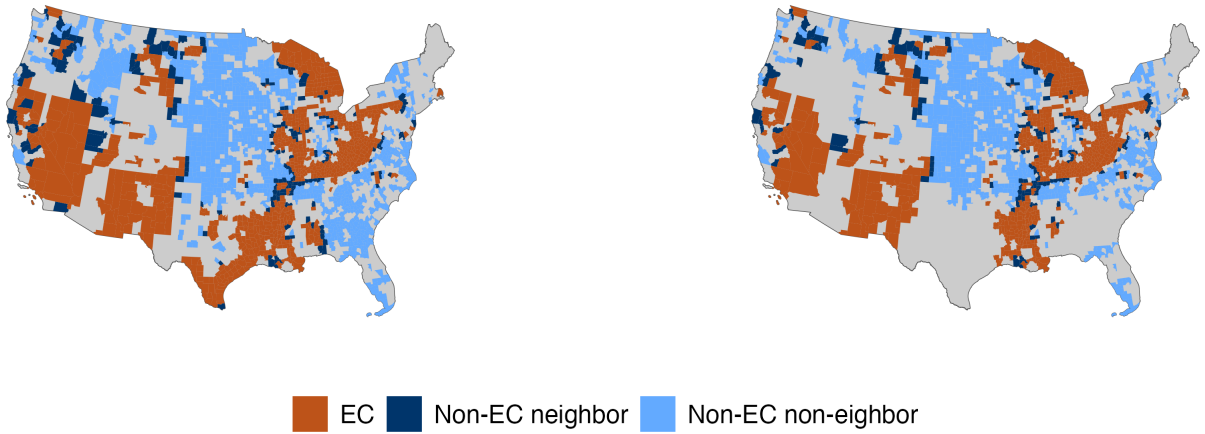
Lastly, I drop counties that neighbor Energy Communities from the set of potential control counties for the empirical analyses. All the analyses in this section identify treatment effects by comparing outcomes in Energy Communities to outcomes in non-Energy Communities. One key concern with this identification strategy is that an exogenous increase in electricity capacity in subsidized areas could causally change capacity in unsubsidized areas. This would violate the stable unit treatment value assumption (SUTVA) required for any causal strategy with a “treatment” and a “control.” In Section 3.4 I describe this concern in more detail and address it with theory and data. For now, I just mention that there is reason to believe that at least one mechanism of spillovers is more likely to

affect neighboring counties, hence I drop them.

3.2 Subsidy Effect on Capacity Additions and Queue Entry

3.2.1 Descriptive Statistics and Empirical Challenges

Table 2 presents descriptive statistics to compare the Energy Community county estimation sample to the rest of the US and the donor pool of non-Energy Community control counties. Figure 2 shows the county locations. Given Energy Communities are chosen to achieve distributional goals, it would be reasonable to suspect they are not a representative sample of US counties. However, Energy Communities are widely dispersed throughout the US, and turn out to be an approximately representative sample of the rest of the US, at least with respect to some key variables relevant to building a new electricity generator. Energy Community counties and the full US sample have similar populations, areas suitable for building clean electricity generators, and transmission wires. Energy Communities have somewhat more queue entry over the pre-period, but given the variance of the county-level aggregates, the values are similar. Energy Communities have had a similar level of capacity additions to the wider US in the pre-period.



(a) Capacity additions

(b) Queue entry

Figure 2: Energy Community counties and the pool of donor controls.

The main empirical challenge is that the donor pool of potential control counties is different, on average, from Energy Communities and the US. The control counties have smaller populations and less transmission than the Energy Community counties, on average. They have similar levels of wind

queue entry and capacity additions, but somewhat less solar and battery. There is a concentration of Energy Communities in the south and southwest, but fewer controls in this area.

The treatment and potential controls also differ in their shares managed by each grid operator. This also poses an empirical challenge for two reasons. First, grid operators have different speeds of processing queues. Second, grid operators collect queue entries in “clusters” of applicants, and have fixed schedules of processing clusters. If Energy Communities are more common in grid operators with faster queue processing or more post-period clusters, that would overestimate the subsidy effects.

I next describe the strategies and robustness checks to estimate subsidy treatment effects on capacity additions and queue entry, in light of these empirical challenges.

3.2.2 Empirical Specification

I use two empirical model specifications, each with distinct advantages. The first is a two-way fixed effects DID with a grid-operator-by-year fixed effect. In particular, I estimate

$$Y_{jt} = D_{jt}\tau + \alpha_j + \beta_{gt} + \varepsilon_{jt}, \quad (1)$$

where Y_{jt} is either queue entry or capacity additions in county j in time t . D_{jt} equals 1 if county j is an Energy Community and time t is after the Energy Community Bonus was enacted. τ is the average treatment effect on the treated (ATT). The regression includes a grid-operator-by-year fixed effect to account for potential spurious correlations between Energy Community prevalence and grid operator queue processing speed or cluster studies.

The primary specification drops counties that neighbor Energy Communities from the control counties because of the possible spillovers I describe in Section 3.4. In event study specifications, I include all time periods, but in pooled treatment-effect estimation I drop some time periods surrounding the January 2023 policy enactment. In particular, I drop late 2022 from the pooled treatment estimates for queue entry analyses. The Energy Community Bonus was announced in August 2022, but not enacted until January 2023, and the guidance that detailed Energy Community locations was not announced until April 2023. Project developers likely speculated on the locations of Energy Communities before they were clear.

The DID has two identifying assumptions. First, that outcomes in Energy Communities and non-Energy Communities would have trended in parallel in a counterfactual world without the Energy Community Bonus policy. Second, that the policy was not anticipated. Note, the DID does not rely on random assignment or similar levels of the outcome variables. It only requires parallel trends.

Variable	All Counties	Queue Entry Sample		Cap. Add. Sample	
		EC	Non-EC	EC	Non-EC
DNI (std. units)	0 (1)	-0.16 (1.33)	-0.13 (0.69)	-0.06 (1.28)	-0.03 (0.72)
Wind 100m (std. units)	0 (1)	-0.16 (0.97)	0.14 (0.86)	-0.17 (0.94)	0.01 (0.93)
Population (10,000s)	10.54 (33.56)	8.68 (49.4)	4.46 (11.5)	9.23 (49.39)	4.75 (11.3)
Area Developed (mi ²)	31.79 (58.4)	27.88 (67.24)	18.02 (21.4)	30.29 (75.45)	18.56 (22.06)
Area Open or Ag. (mi ²)	252.21 (229.96)	203.01 (204.76)	319.74 (245.35)	212.25 (204.72)	288.98 (242.4)
Area Bare, Grass, Shrub (mi ²)	365.21 (1121.33)	500.45 (1607.14)	196.49 (470.83)	548.1 (1683.37)	222.37 (551.1)
Transmission (kV×km 1,000s)	50.98 (74.26)	50.72 (110.59)	33.24 (34.75)	53.61 (107.73)	35.11 (39.01)
PJM or MISO	0.51	0.8	0.58	0.72	0.47
NYISO or ISONE	0.05	0.01	0.04	0.01	0.03
CAISO or SPP	0.26	0.13	0.41	0.16	0.35
ERCOT	0.08	0	0	0.13	0.02
Non-ISO	0.49	0.39	0.47	0.38	0.55
<i>Queue Entry 2017-2022 (MW)</i>					
Solar	299.52 (697.23)	415.58 (864.71)	197.02 (455.43)		
Wind	149.81 (941.19)	105.24 (556.79)	136.38 (718.62)		
Battery	200.86 (825.11)	270.18 (1236.55)	91.69 (292.21)		
Other Elig.	3.75 (45.06)	4.22 (39.26)	2.24 (40.15)		
<i>Capacity Additions 2017-2022 (MW)</i>					
Solar	16.16 (79.24)			15.43 (111.58)	10.39 (48.83)
Wind	19.42 (92.59)			19.39 (83.73)	19.37 (88.74)
Battery	2.75 (37.06)			5.19 (65.42)	0.99 (19.57)
Other Elig.	1.34 (14.37)			1.3 (12.12)	1.15 (17.6)
N	3,108	621	1,024	739	1,261

Table 2: Descriptive statistics and balance table. Means with SDs in parentheses, except the grid operator measures, which are shares. Shares sum to more than 1 because some counties are covered by more than one grid operator. “All Counties” queue entry measures use the 2,558 counties with queue data. EC and non-EC counties sum to less than 3,108 due to dropping partially-treated counties and, for the queue sample only, missing data. DNI is direct normal irradiance, or sunniness. DNI and wind speed are measured as county maximums.

The second empirical model I use is a synthetic DID, following the method of Arkhangelsky et al. (2021):

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min_{\tau, \mu, \alpha, \beta} \left\{ \sum_j \sum_t (\tilde{Y}_{jt} - \mu - \alpha_j - \beta_t - D_{jt}\tau)^2 \hat{\omega}_j \hat{\lambda}_t \right\}. \quad (2)$$

The equation is the same as the simple two-way fixed effects DID specification with a few key amendments. First, the estimator chooses unit-weights $\hat{\omega}_j$ to minimize the difference between the pre-period trends between the treated control units. Second, time-weights $\hat{\lambda}_t$ are chosen to select the pre-treatment periods that best match the post-treatment periods, such that the pre-period difference is not driven by outlier periods. More detail on calculating the weights is in the appendix. μ is an intercept to allow for the parallel difference between the treatment and control trends.

The outcome variable is denoted $\tilde{Y}_{jt}$ with a tilde due to a technical detail that enables including the grid-operator-by-year fixed effect. I include the fixed effect following the method in Kranz (2022). In particular, I estimate the linear model

$$Y_{jt} = \alpha_{gt} + \gamma_t + \mu_j + \varepsilon_{jt}.$$

I estimate the model using all counties in the pre-period, and only control counties in the post-period. This way, α_{gt} is the average effect on Y_{jt} of being located in grid operator g 's jurisdiction in year t , not including the treatment effects. The equation is not simply a fixed effect because some counties are located the jurisdictions of multiple grid operators. Finally, I calculate the adjusted $\tilde{Y}_{jt} = Y_{jt} - \sum_g g_j 1\{t = s\} \hat{\beta}_{gs}$ for all treatment and control units. Including this first stage requires bootstrapping to calculate standard errors. The bootstrap procedure is similar to Arkhangelsky et al. (2021) Algorithm 2, and follows the guidance of Kranz (2022).

Another option for empirical approach is to use a regression discontinuity design (RDD) on the fossil fuel employment definition of Energy Communities, which includes two sharp thresholds. However, RDD assumes the outcome variable is a continuous function of the running variable around the threshold. This is not the case with electricity capacity additions and fossil fuel employment rates, so the RDD produces imprecise results. While including many controls improves precision, this changes the interpretation of the estimand and the precision improvement turns out to be minor.

3.2.3 Queue Entry and Short-run Capacity Addition Results

Figure 3 shows the TWFE DID results from the simple specification in Equation 1. The queue-entry regressions are at the half-year unit of analysis because the data ends after the repeal announcement in July 2025, while the capacity addition regressions are at the annual level. The capacity

addition regressions are measured in megawatts (MW) per county per year, and the queue entry regressions in MW per county per half-year.

On the capacity additions results, the Energy Communities and control counties trend approximately in parallel in the pre-period, even with the simple DID strategy. The short run capacity addition effects are significant at a less than one percent p-value for solar and all eligible MW. They are insignificant for batteries and wind. The battery insignificance could be explained by the fact that there are relatively few grid-scale batteries coming online during this time, compared to the many batteries in interconnection queues. Any significant effects do not begin until 2024, a year after the policy enactment.

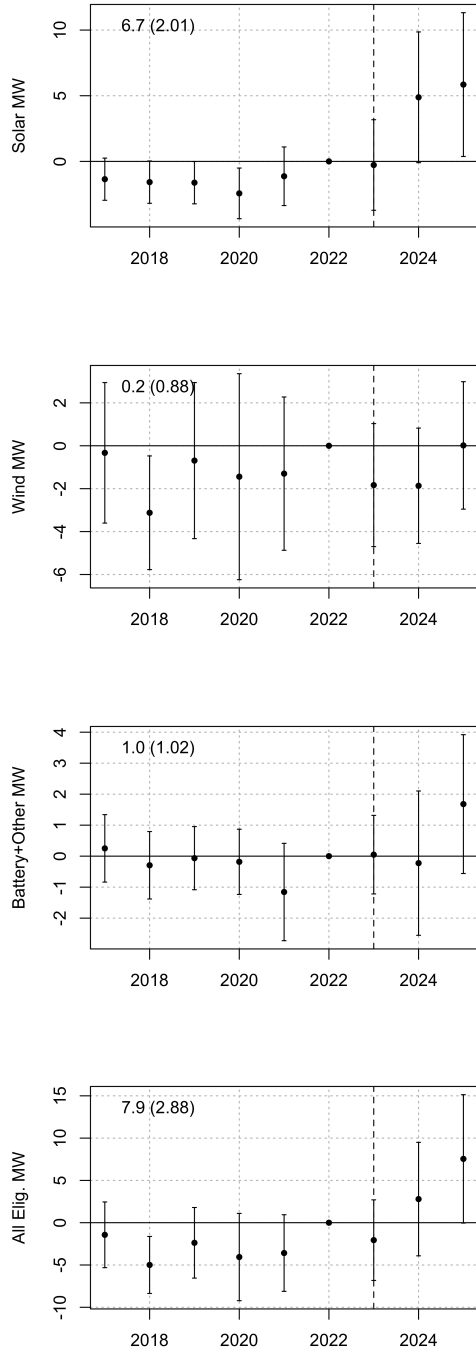
On queue entry, results are positive and significant for all technologies. The drop in estimated treatment effect in the second half of 2024 is because that time period happened to not align with many cluster studies. The largest effects are for solar and batteries. The solar and battery regressions are estimated more precisely than the wind regressions because there are fewer zero-valued observations. There may be slight pre-trends in solar and battery queue additions, but if so, they are not extreme. Though not significant, under pre-trends, the SDID should better eliminate this bias by choosing control counties with pre-trends parallel to the Energy Communities.

Figure 4 reports the SDID results for queue entry and Figure 5 for capacity additions. The top row shows the mean trends for the Energy Communities and a synthetic control group. The bottom row shows a permutation test, generated by re-running specification in Equation 2 1,000 times, and with each permutation re-sampling the counties with replacement and randomly assigning treatment status. For both queue entry and capacity addition outcomes, solar, battery, and pooled estimates are positive and statistically significant.

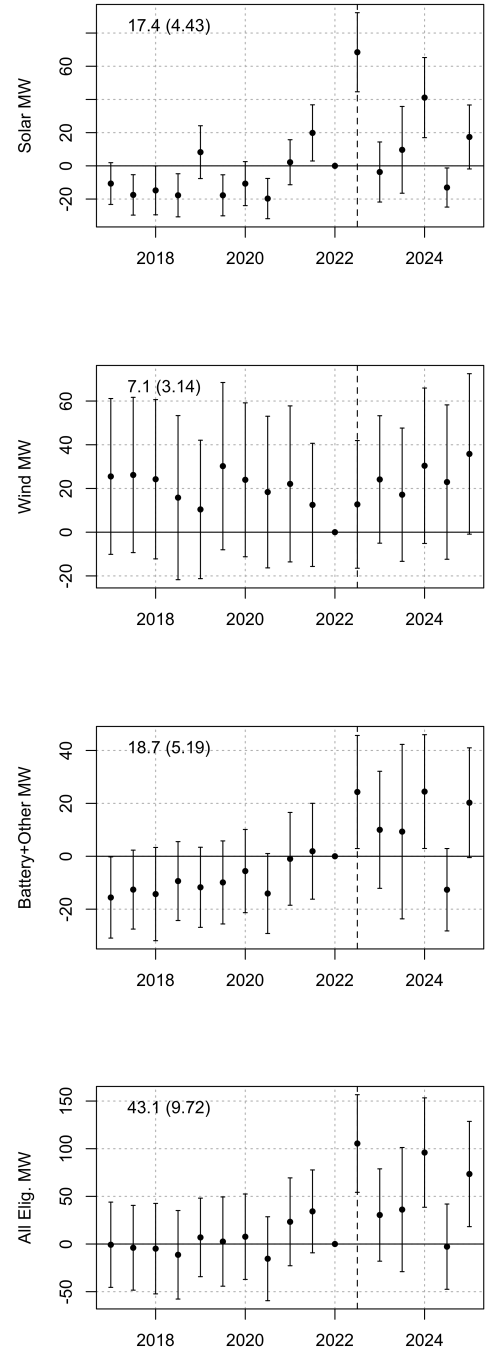
The effect sizes for the SDID are similar to the DID regressions, except for battery capacity additions, where they are larger. For a sense of magnitude of the capacity addition result, compared to a counterfactual without the treatment effect, the treatment effect suggests a doubling of solar capacity additions and a five-times multiplier on battery capacity additions. The queue entry results are a 70 percent increase for solar and more than a doubling for batteries.

3.2.4 Heterogeneity and Robustness Checks

Appendix B reports heterogeneity by grid operator. This exercise still requires some aggregation because certain grid operators do not have a sufficient number of Energy Community or non-Energy Community controls. The results tend to lose significance with the disaggregation, though some



(a) Capacity additions



(b) Queue entry

Figure 3: Event studies, with grid-operator-by-year fixed effect. Pooled treatment effect at the top-left with standard errors clustered at the county level.

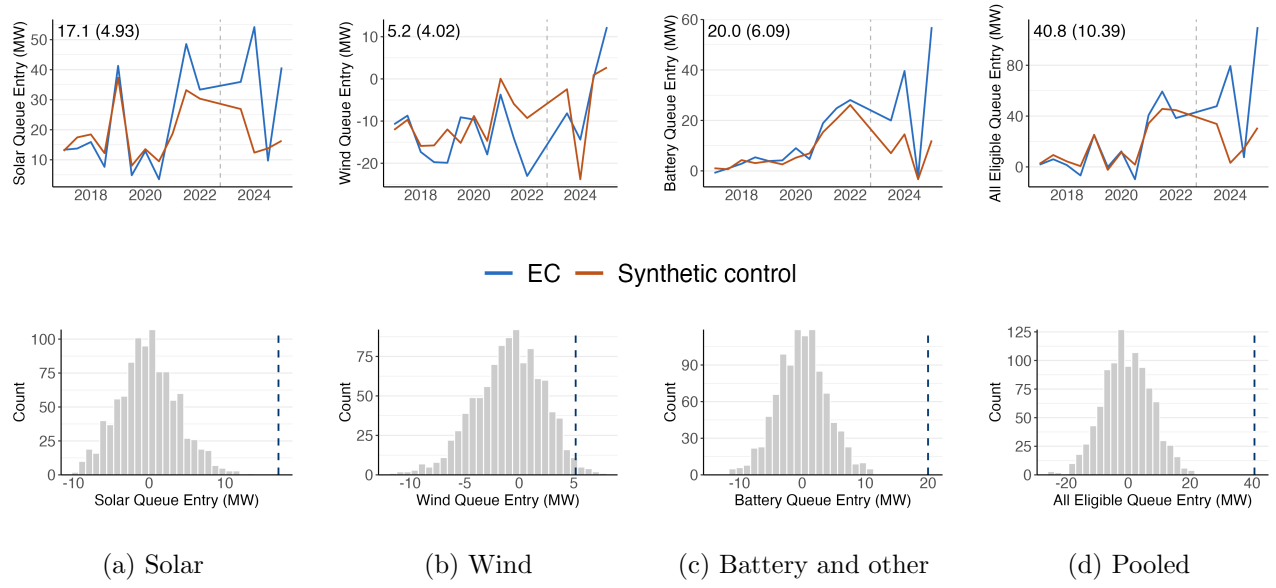


Figure 4: Queue entry SDID and permutation test results. Pooled treatment effects reported at the top, with bootstrapped standard errors in parentheses.

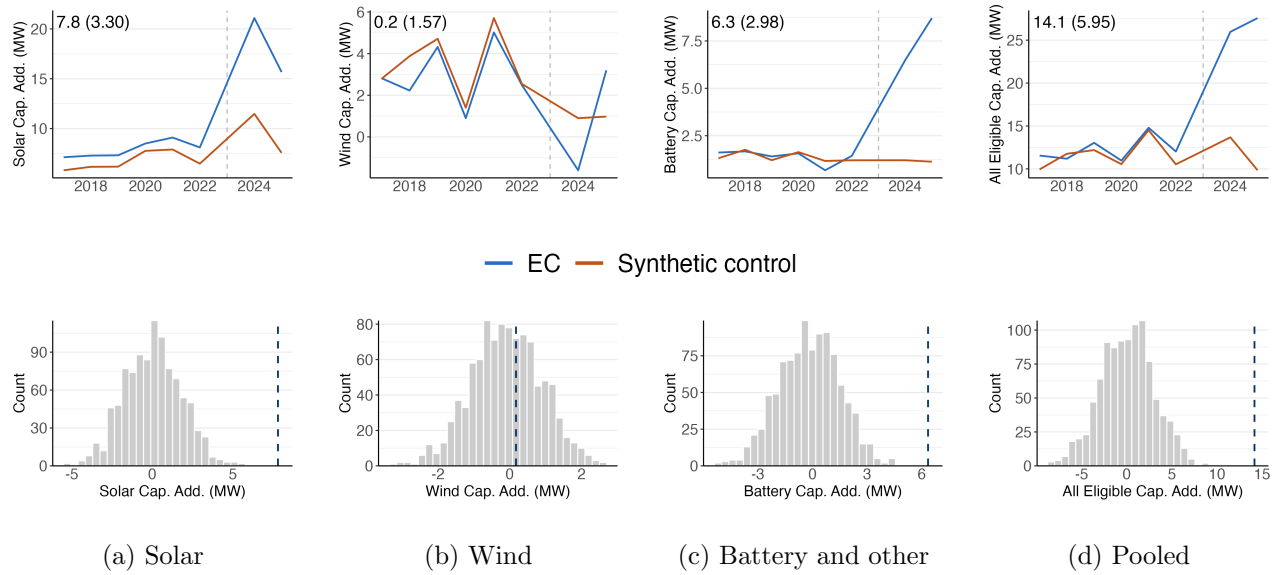


Figure 5: Capacity addition SDID and permutation test results. Pooled treatment effects reported at the top, with jackknife standard errors in parentheses.

remain significant. MISO, SPP, CAISO, ERCOT, and the non-ISO regions have the same qualitative takeaways upon disaggregation. PJM, NYISO, and ISO-NE show null or even insignificantly negative effects.

One concern highlighted by the maps showing Energy Community counties and non-Energy Community controls (Figure 2) is that the Energy Community counties tend to be in the sunnier west or southwest. Adding to that concern, given the prevalence of counties with zero capacity additions or queue entry, the unit fixed effects may not effectively control for geography, and the SDID algorithm may not effectively select controls accounting for geography.

The DID pre-trends being approximately parallel largely quells this concern for the main specification. But to further interrogate the result, in Appendix C presents three more SDID analyses. The first drops Arizona, California, Nevada, and New Mexico. Appendix Figure 12 shows the results. Queue entry results for solar, batteries, and aggregate are all still positive and statistically significant. The coefficients are smaller than with keeping those states, but still quite large. Capacity addition results are no longer significant, but are all positive and large.

The second robustness adds a state-by-year fixed effect to the baseline specification, which effectively drops Arizona, Nevada, and New Mexico because there is no variation in Energy Community status in those states' sampled counties. The third keeps the state-by-year fixed effect and also drops California. I caveat that since states have varying numbers of Energy Community counties, including a state-by-year fixed effect eliminates a valid channel of treatment effect. If the subsidy has an effect, states with more Energy Communities should see larger average capacity addition or queue entry changes in the post-period, but the state-by-year fixed effect eliminates that channel. For this reason, the state-by-year fixed effect is an intentional omission from the primary specification. However, given all states have some capacity additions and queue entry in every year, it effectively controls for trends in geography.

Appendix Figure 13 shows the result of including the fixed effect, and Appendix Figure 14 also drops California. The standard errors are much larger. All the coefficients are large and positive. Some lose significance, but solar capacity additions, solar queue entry, battery queue entry, and aggregate queue entry all retain significance even with the fixed effect. Largely these results confirm the sunny western states do not exclusively drive the estimates.

Taken together, the analyses suggest more capacity additions, even when generators do not endogenously choose to locate there, and more queue entry in Energy Communities due to the subsidy. The next section measures outcomes for the projects entering the queue with the subsidy.

3.3 Interconnection Costs and Queue Completion Probability

As I described in Section 1.2, it is straightforward to hypothesize that the Energy Community Bonus capitalizes into projects' WTA for their interconnection costs. If this is the case, projects in Energy Communities should have a weakly higher WTA for interconnection costs than projects in comparable non-Energy Communities. The inframarginal queue entrants receiving the subsidy—those that would have entered the queue regardless of the Bonus—have a higher WTA exactly equal to the Bonus value. The marginal queue entrants—those that would not have entered the queue without the promise of the Energy Community Bonus, or the projects comprising the last section's treatment effect—have a WTA between zero and the Bonus value.

The first empirical analysis in this section tests this hypothesis. The value of that exercise is to empirically assess the Energy Community Bonus value and a mechanism by which it could increase new capacity additions.

However, the ultimate question of this paper is how a marginal subsidy affects Energy Community capacity additions in the long run. The second empirical analysis in this section is toward this end. It estimates the probability projects in Energy Communities accept their interconnection costs, or their “survival” probability. Unlike the case of interconnection costs, which has a clearly signed hypothesis, the survival probability of projects in Energy Communities compared to projects in comparable non-Energy Communities is ambiguous. The inframarginal queue entrants in Energy Communities should have a higher survival probability than comparable projects in non-Energy Communities. However, marginal queue entrants should have a lower survival probability than inframarginal entrants, and an ambiguous survival probability compared to projects that do not receive the subsidy. Therefore, the direction, let alone the magnitude, of Energy Community projects' survival probability compared to a counterfactual survival without the subsidy is an empirical question.

Before estimating the effect of the subsidy on costs and survival probability, I describe the available data and two points of empirical analysis that require special attention.

3.3.1 Descriptive Statistics and Empirical Challenges

Table 3 presents descriptive statistics of the data from MISO, SPP, and PJM used in this section. The left-most column shows the descriptives of the full dataset. The right two columns show Energy Community and non-Energy Community descriptives.

The mean quoted cost to connect to the grid from the first study is nearly \$50 million, or \$345 thousand per MW. The standard deviation is larger than the mean, underscoring the high variability

in costs, and suggestive of the uncertainty. Non-first study costs are lower on average, reflecting the tendency of projects to reject studies with high costs. Even more suggestive of the cost uncertainty is that the standard deviation of the change in costs between studies is nearly as large as the mean cost.

One empirical challenge is related to censoring bias and the short time horizon between policy enactment and repeal. The second section of the Table 3 shows, by entry cohort, the share of projects that are censored or, in other words, do not have an observable outcome. An observable outcome is either that the project withdrew pre-study, or has had at least one study. It is problematic to selectively use the uncensored projects for estimation because projects may select into early outcome arrivals. For example, particularly valuable projects may have a study arrive earlier, or particularly unprofitable projects may withdraw the earliest.

The forthcoming analyses are DID. Censoring is a non-issue for projects that enter the queue in the pre-period, as about 85 percent of them are uncensored. This masks heterogeneity by grid operator, and the value is over 90 percent for SPP and MISO. Censoring is a larger issue in the post-period: 49 percent of the 2023 cohort of queue entrants are uncensored. This again masks grid operator heterogeneity: exactly zero PJM projects that entered the queue in 2023 have had a study within the data sample, compared to 92 percent of MISO or SPP projects. Under all grid operators, 2024 and 2025 queue entrants are highly censored, with only about 10 percent of projects reaching a first outcome.

The other empirical challenge concerns the fact that the Energy Community Bonus causes marginal entry, and the outcomes of interest being for inframarginal and marginal entrants combined. Although there have been over 2,000 cost studies in Energy Communities and non-Energy Communities between 2023 and June 2025, most of the studies are of projects that entered the queue without endogenously accounting for the Bonus. While still of interest, the interpretation of the outcomes for these projects does not represent a long-run treatment effect in which projects endogenously account for the Bonus in their entry decision.

The next section analyzes how the Bonus capitalizes into interconnection costs. For this metric, I keep the censored projects and inframarginal entrants, as it is simply a descriptive valuation of the subsidy. The following section on survival probability more carefully considers censoring and inframarginal projects in its empirical specification.

Variable	All Counties	All EC	Non-EC
N projects entering queue			
in pre-IRA	6,934	2,252	1,628
in 2023	874	215	232
in 2023 (w.o. PJM)	426	73	152
in 2024-25	1,200	519	241
Share studied or withdrawn pre-study			
pre-IRA entrants	0.85	0.85	0.87
2023 entrants	0.49	0.33	0.63
2023 entrants (w.o. PJM)	0.92	0.86	0.94
2024-25 entrants	0.09	0.10	0.07
N studies			
of pre-IRA entrants, pre-IRA	3,069	852	889
of pre-IRA entrants, 2023 or later	3,872	1,305	1,047
of 2023 entrants	716	133	228
of 2024-25 entrants	58	23	13
Initial study cost (\$M)	49.2 (79.5)	61.8 (92.0)	41.2 (61.7)
Initial study cost per MW (\$M)	0.345 (0.539)	0.421 (0.614)	0.299 (0.439)
Non-first study cost (\$M)	41.0 (67.3)	57.0 (82.5)	31.0 (48.1)
Non-first study cost per MW (\$M)	0.251 (0.338)	0.344 (0.383)	0.208 (0.292)
Change from prior study (\$M)	-1.1 (44.2)	-1.5 (51.8)	-1.0 (39.7)
Change from prior study per MW (\$M)	-0.017 (0.334)	-0.018 (0.307)	-0.011 (0.307)
Share of withdrawn that withdrew pre-study or within 3 mo. of last study	0.89	0.90	0.91
Project technology shares			
Hybrid	0.12	0.15	0.09
Solar or wind	0.70	0.70	0.72
Battery	0.19	0.14	0.19
Capacity shares			
Battery	0.20	0.19	0.18
Solar	0.60	0.70	0.51
Wind	0.19	0.11	0.31
Project shares			
MISO	0.47	0.64	0.33
SPP	0.24	0.11	0.37
PJM	0.29	0.26	0.29

Table 3: Summary statistics for the interconnection cost data. Other than the counts and shares, the numbers are means with standard deviations in parentheses.

3.3.2 Interconnection Cost Effects

Figure 6 shows the interconnection costs per MW for projects in MISO, SPP, and PJM. The leftmost panel shows the distribution of costs for withdrawn and not withdrawn projects, separated by technology. Withdrawing after a study is defined as withdrawing within 3 months of the study. Not withdrawing is defined as not withdrawing within 3 months. Unsurprisingly, projects that have high costs tend to withdraw, and those that have low costs tend to not withdraw.

The middle and right panels show accepted costs by technology in Energy Communities and non-Energy Communities. The middle panel is pre-policy data, and the right panel is post-policy data. Before the policy, the mean costs in Energy Communities and non-Energy Communities are the same. After the policy, projects in Energy Communities accept higher costs. Appendix Figure 15 shows these plots as event studies, estimating the equation,

$$ICCost_{ist} = \tau_c D_{ist} + \gamma EC_i + \alpha_{gt} + \varepsilon_{ist},$$

where D_{ist} is one if study s of project i is in an Energy Community in 2023 or later and zero otherwise, and EC_i is one if project i is in an Energy Community. There is no unit fixed effect because the data are largely cross-sectional; only a few projects have studies before and after the policy. $ICCost_{ist}$ is the interconnection cost. Like the box-plots, the sample includes only cost offers after which projects do not withdraw. τ_c is the estimated subsidy capitalization into interconnection cost WTA.

The pooled effect yields a mean difference of about \$142 thousand per MW. The annual cost data are noisy, but pre-trends are approximately parallel and the figures have the same takeaway. In calculations later in the paper, I show the subsidy has a present discounted private value of about \$145 thousand per MW for solar, \$170 thousand per MW for batteries, and \$80 thousand per MW for wind, with substantial variation by technology and location. It would make sense that the difference in accepted costs is less than the value of the subsidy, given not every project is the marginal project that capitalizes the full value of the subsidy into its interconnection costs. So, this mean capitalization seems slightly high based on my own calculations, but within a sensible range and is evidence the Bonus capitalizes into interconnection cost WTA.

Projects accepting higher costs of course does not necessarily imply they have a higher probability of acceptance. Perhaps they simply face higher costs. I next turn to survival probability estimation.

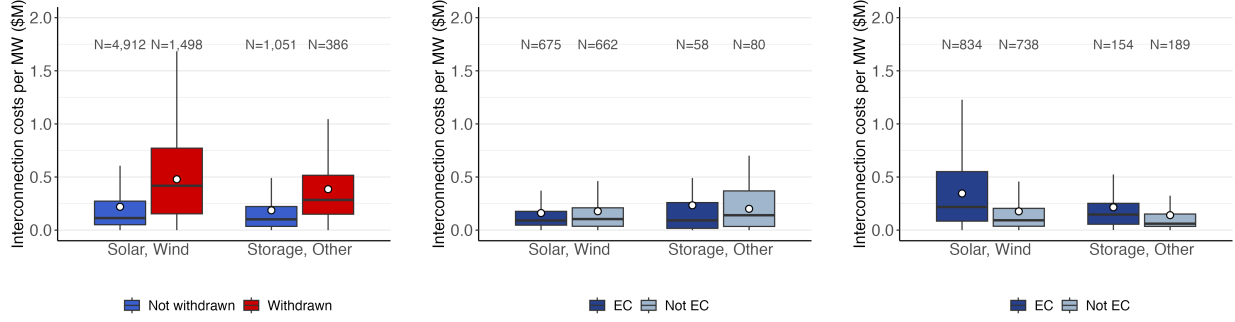


Figure 6: Interconnection costs per MW by technology and EC type in real 2025 USD. Hybrid generation and storage projects are grouped with solar and wind. The boxes show the 25th and 75th percentiles, the black line the median, the white dot the mean, and the whiskers 1.5 times the interquartile range up to the minimum or maximum.

3.3.3 Interconnection Cost Acceptance Probability Effects

I estimate the causal effect of the Energy Community Bonus on Energy Community projects' probability of withdrawal from the queue with the equation,

$$W_{ist} = \tau_p D_{it} + \gamma EC_i + \alpha_{gt} + \varepsilon_{ist} \quad (3)$$

where W_{ist} equals one if project i withdraws from the queue after study phase s in time period t and zero otherwise; τ_p is the reduction in the probability of project withdrawal due to the subsidy. Like the queue entry regressions, the pooled estimates drop projects that entered the queue after the Energy Community Bonus was announced in August 2022 but before the locations of Energy Communities were clear in April 2023. I also drop projects in counties that neighbor Energy Communities that may be affected by the Energy Community Bonus via spillover effects (more on this shortly).

I estimate withdrawal at two study phases. The first is pre-study withdrawal and initial studies combined, and the second is all non-first studies combined. The first regression is at the project level, since survival through the first study is a project-level outcome. The second regression is at the study level, since projects may have multiple non-first studies. These two regressions require different definitions of time units t . It is perhaps easier to explain this while going through the results, so I defer the explanation until then.

I limit the estimating sample to MISO and SPP, dropping PJM. I also truncate the post-period sample to include only 2023 queue entrants, dropping 2024 and 2025 entrants. These two adjustments limit censoring bias, as about 90 percent of these post-policy entrants are uncensored, which is similar

to the pre-policy rate. In MISO and SPP, of the approximately 200 projects that entered the queue and had a study, over 99 percent of them have had a second study. Therefore, the non-first study outcome should also be little affected by censoring bias.

My main specification drops projects and studies for projects that enter the queue before the policy, but do not have an outcome until after the policy. I also run the analyses keeping these projects, with the caveat the interpretation of the estimate is not a long-run effect.

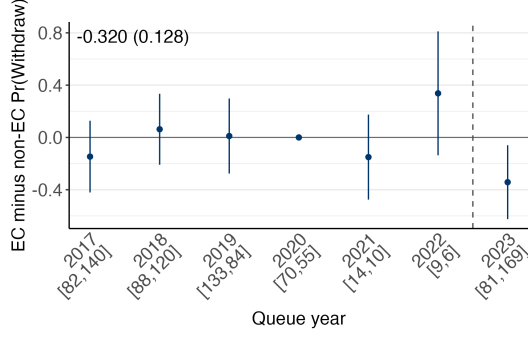
The downside of these approaches to censoring bias and estimand interpretation is that it limits statistical power. It turns out to yield sufficient power for analyses that aggregate all technology types, but proves overly limiting for certain technology-level regressions.

Perhaps notably, I do not control for interconnection costs. This is a “bad control” in that it is also affected by the subsidy. Controlling for interconnection costs would yield the withdrawal probability effect omitting whatever role is contributed by the higher costs shown in the previous section.

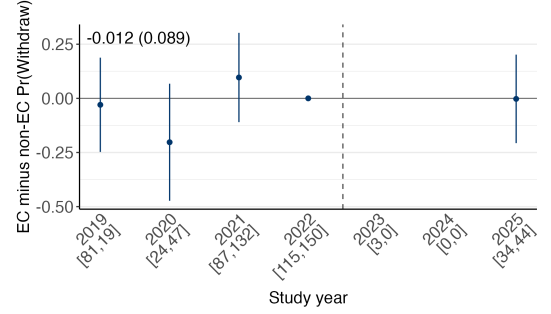
Another popular option to handle censoring bias is a duration model (Abbring and Van den Berg, 2003). In principle, I could also ignore cost studies altogether, use withdrawal timing data from all grid operators, and estimate a duration model that predicts project withdrawal probability. However, these models would not debias the regressions because the censoring is, to an extent, a variable that projects can control as they select into receiving an early study or early withdrawal. Furthermore, the outcome would be measured in a duration, and I would need to make some parametric assumption to convert the estimate to a survival probability. Instead, I prefer the simple linear model. The economics literature on crime recidivism and unemployment policies takes a similar approach (e.g., Rose (2021); Lichter and Schiprowski (2021)).

Figure 7 shows the results of estimating Equation 3 as an event study, with the pooled estimate and county-clustered standard errors at the top left. The left panel shows estimates of the subsidy effect on the probability a project withdraws pre-study or after the first study. This regression is at the project level, and the x-axis tracks the queue entry cohort of the projects. Queue entry cohort is sufficient to separate the pre- and post-policy periods because the regression drops the projects that entered the queue pre-policy and have a pre-study withdrawal or initial study outcome after the policy.

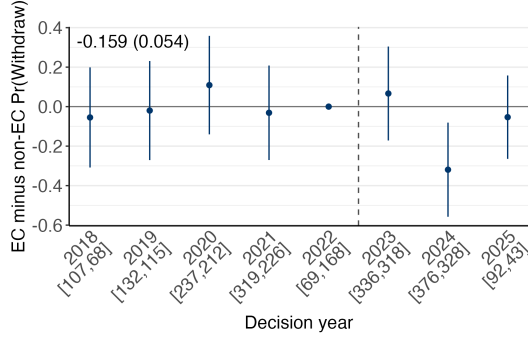
The Energy Community Bonus causes a substantial reduction in the probability of withdrawal. Trends are parallel before the policy, with some noise in the pre-trends for queue cohorts that are sparsely covered due to dropping pre-policy entrants with a post-policy outcome. Projects in Energy Communities after the policy are 32 percent less likely to withdraw before their study or after their first study. This is the combined result for inframarginal and marginal Energy Community queue



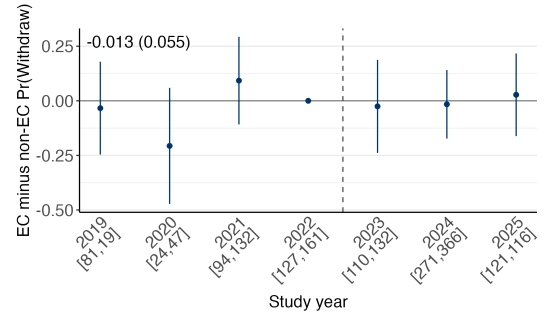
(a) Initial study, drop pre-IRA entrants



(b) Non-first studies, drop pre-IRA entrants



(c) Initial study, keep pre-IRA entrants



(d) Non-first studies, keep pre-IRA entrants

Figure 7: Event studies of how the subsidy affects projects' probability of withdrawing from the inter-connection queue. Regressions include time fixed effects (queue year or study year) and a treatment indicator. Capacity-weighted. The pooled coefficient estimate is at the top left, with standard errors in parentheses, clustered at the county level. The numbers in brackets below the years show the number of projects underlying the observation in Energy Communities and non-Energy Communities, respectively.

entrants.

The right panel shows the subsidy effect estimates on withdrawal after non-first studies. This regression is at the study level, unlike the previous one, which is at the project level. So, the x-axis is tracking the year of the non-first study. No projects in non-Energy Communities entered the queue in 2023 and had a non-first study before 2025, hence the empty estimates. However, there are 80 studies to compare outcomes for, and they suggest the subsidy has little or no effect on survival rates after the initial study period.

The bottom row of Figure 7 shows these same results, but retaining the inframarginal entrants. The time unit in panel 7c is the decision year. This is either the year that the project withdraws

pre-study, or its initial study year. For the non-first studies, the time unit is simply the study year. Like the queue entry regressions above, the pooled estimate drops decisions in the second half of 2022 and first half of 2023, since during this time, the Energy Community Bonus had been announced but the locations of Energy Communities were unclear.

Retaining the pre-IRA entrants yields an estimate that indicates significantly higher survival in Energy Communities through the first study, and no effect through non-first studies. The perhaps surprising feature is that the effect is smaller when dropping the pre-IRA entrants. However, the coefficients are not statistically significantly different.

Taken together, the previous analyses suggest large long-run effects of the Energy Community Bonus because projects are not only more likely to enter the queue in Energy Communities, but also more likely to survive the queue process and become operational. I calculate the implied long-run capacity addition effect in Section 3.5. All of the empirical analyses in this paper estimate treatment effects by comparing Energy Communities to non-Energy Communities. The next section evaluates the possibility that subsidizing Energy Communities affects projects in non-Energy Communities, which would bias the comparisons.

3.4 Spillover Effects

A concern with any empirical strategy using spatial variation for identification is that the treatment affects the control group, violating SUTVA. In the context of this clean electricity subsidy, there are three plausible mechanisms that could facilitate a SUTVA violation. The first is about grid infrastructure costs, the second grid congestion, and the third traditional economic price mechanisms.

On infrastructure costs, building a new clean electricity generator may affect the cost to build other projects. This is an externality, and its value can be positive or negative. It can be positive because projects in the queue often must share the costs of grid infrastructure, so the presence of a project can lower costs for others. It can be negative because the presence of a project can increase grid infrastructure requirements for other projects.

The grid congestion spillover effect is another externality that could be positive or negative. Building a new project could require building a new transmission line, which could lower grid congestion for other projects. Building a new project could also congest planned or existing lines, lowering the values of other projects. For batteries, grid congestion is a benefit, not a cost. Grid congestion lowers the marginal value of electricity on a transmission line to zero, creating an arbitrage opportunity. So, while batteries are still exposed to these externalities, the physical effects are associated with the

opposite externality signs.

Finally, the traditional economic price mechanism is that subsidizing electricity in one area (weakly) lowers the equilibrium price of electricity in all areas. Intuitively, the subsidy shifts the marginal cost curve downward, and lowers the price. This effect could materialize in markets for electric energy, as well as markets for electric capacity reliability. Section E.2 presents a formal model of this, but I delay the theory-informed discussion until after an empirical evaluation of local spatial spillovers.

3.4.1 Local Spatial Spillovers

Any of these externalities may occur in nearby counties or distant counties. However, I speculate the two infrastructure-based externalities are more likely to impose an externality on a nearby project than on any given distant project. A project may trigger new grid infrastructure or congestion anywhere, but it is more likely to do so where it is located.

I test the net direction of the infrastructure externality by re-estimating all the analyses above, dropping “treated” Energy Communities, and instead comparing counties that neighbor Energy Communities as the “treatment” group to further away counties (Butts, 2023; Clarke, 2017). I only use neighboring counties that meet none of the Energy Community definitions in any year, and so would otherwise be in the control group if not for neighboring Energy Communities. The control groups in these analyses are the same as before because I excluded the neighbors from the previous control groups. With these samples, I test whether the Energy Community Bonus has a detectable causal effect on capacity in nearby counties.¹³

Appendix E.1 reports the results. First, I re-estimate the DID in 1 and SDID in Equation 2 for the recent capacity addition and queue entry outcome variables, with results in Appendix Figures 17 and 18. For capacity addition results aggregating MW for all eligible technologies, the coefficients are positive and significant. The capacity addition results are driven by solar and batteries. For queue entry, the results are all relatively precise nulls, especially for solar and wind. The battery SDID coefficient is somewhat larger, but still null. This suggests in the short-run, if there is any spatial spillover effect, it is a crowd-in. In the long run, I interpret the results as null, despite the positive values.

Appendix Figures 19 and 20 show the re-estimated interconnection cost results, with projects in

¹³A possible concern with this approach is that the neighbors have more Energy Community brownfields than the distant controls, given I observe brownfields imperfectly. For the brownfields I do observe, the neighbors and second-and-farther neighbors both average 2 brownfields per county, while the treated Energy Communities average 11 brownfields per county. So, as far as is possible to tell with the available data, this is a non-issue.

Energy Community neighbors as the treated group and projects in more distant counties as controls. Projects in counties that neighbor Energy Communities accept higher interconnection costs. Their first study survival rate differences compared to the non-neighbor counties are not significantly different. Unfortunately, I do not have enough statistical power to measure non-first study outcomes for the neighbor group. There are only two projects that meet the criteria for treatment. Keeping the pre-IRA entrants shows null results. This is in contrast to the Energy Community projects receiving the subsidy, which face similarly higher costs but have lower withdrawal probability.

Taken together, these results suggest that in the short run, a marginal subsidy crowds-in nearby solar and storage capacity, perhaps due to the cost-splitting mechanism. There are no long-run spillovers, after location is endogenous, which accounts for the influx of marginal queue entrants within Energy Communities.

One takeaway from these results could be a partial explanation for the very large results in the analyses of Energy Community Bonus effects on Energy Communities. The inter-project externalities occur within the subsidized areas. Thus, the mass of queue entry in Energy Communities not only creates competition, but also allows other projects to reliably split interconnection costs. In the case that grid infrastructure is a club good, there are conditions in which a subsidy could crowd-in investment.

These empirical results capture local spillovers, but spillover effects do not necessarily have to be localized. Perhaps spillovers contaminate the entire control group. In principle, that bias could go in either direction: the empirical results could overstate subsidy effects if the Energy Community Bonus crowds out electricity capacity in distant counties, or could understate subsidy effects if distant capacity also experiences crowd-in effects. This is challenging to measure empirically because there is no suitable comparison group for treatment effect estimates. Instead, I argue any distant spillover effects and the resultant estimate biases are likely small.

3.4.2 Distant Spillovers

In principle, a new generator *could* cause additional infrastructure or grid congestion *anywhere* on the interconnected grid. But it surely will not cause additional infrastructure and congestion *everywhere* on the grid. That is to say, most of the control group that does not neighbor Energy Communities is unlikely to be as contaminated as the neighbors.

On the other hand, price effects are likely to occur everywhere on the grid. Although many balancing authorities feature “nodal marginal prices” (i.e., local pricing), which suggest localized price

effects, the grid is not usually completely congested and electricity is traded across grid operators, so price effects could have far-reaching spillovers. Nonetheless, I argue any price effects induced by this marginal subsidy are small.

Appendix E.2 presents a simple general equilibrium model of an electricity market in the long run. The model applies to either electric energy or capacity markets. There is aggregate demand for electricity $x(p)$. Every county j offers supply of subsidy-eligible new electricity $y_j^a(p)$ (a for addition), supply of clean electricity pre-existing generators $y_j^b(p)$ (b for baseline), and supply of new and pre-existing fossil fuels $f_j^a(p)$ and $f_j^b(p)$.

Consider two comparable counties, $j = E$ an Energy Community and $j = N$ a non-Energy Community. The comparability comes from, for example, the SDID procedure. Under a price-driven spillover effect, comparing the Energy Community to the non-Energy Community would over-estimate the treatment effect by a factor,

$$\frac{dy_E^a - dy_N^a}{dy_E^a} = 1 - \frac{\sum_j \theta_{y_j^a} EC_j \varepsilon_{y_j}}{\varepsilon_x - \sum_j \theta_{y_j^a} (1 - EC_j) \varepsilon_{y_j^a} - \theta_{f^a} \varepsilon_{f^a} - \theta_{y^b} \varepsilon_{y^b} - \theta_{f^b} \varepsilon_{f^b}}, \quad (4)$$

where ε indicates the price elasticity of respective demand or supply according to the subscript, dropping the subscript j indicates an aggregate elasticity, θ indicates the share of the total quantity of electricity supply in equilibrium from the subscripted source, and EC_j is an indicator equal to one if county j is an Energy Community and zero otherwise.

These parameters are challenging to estimate in practice, largely because the price p is some aggregator of present prices and developers' expected future prices.¹⁴ However, reasonable assessment suggests the multiplier is close to one.

First and foremost, although my results suggest large effects of the Energy Community Bonus on capacity additions, the margin of capacity additions is small relative to the entire US electricity grid. The multiplier is close to one if the share of total electricity coming from new generators in Energy Communities is small. Supposing homogenous counties for simplicity, this is true if $\sum_j \theta_{y_j^a} EC_j$ is small. My estimates below suggest the Energy Community Bonus causes about 40 GW of new clean electricity capacity added to the grid per year. This is a lot, but even if repeated for many years, it is small relative to the full grid, which had about 1,280 GW of nameplate capacity in 2025. Also, 1,280 GW does not account for future baseline capacity additions in Energy Communities and elsewhere. Furthermore, these numbers are nameplate capacities, so they do not account for capacity factors.

¹⁴If one could come up with a measure of price expectations, the Energy Community Bonus would provide sufficient variation as an instrument to estimate both the supply and demand elasticities, using the framework in Zoutman et al. (2018). I leave this for future work, presenting this model for argument's sake.

The intermittent renewable generators and batteries on the grid should have lower capacity factors than the existing fleet including its fossil fuel generators, which suggests the new renewable generators are an even smaller share of generation. Regardless of the elasticity parameters, the spillover effect should be multiplied by some single-digit percent $\sum_j \theta_{y_j^a} EC_j$.

Considering the elasticity parameters, some of these also suggest the multiplier is close to one. The multiplier is close to one if long-run electricity demand is elastic (ε_x is very negative). There is limited evidence of long-run electricity demand elasticity, but Buchsbaum (2022) finds a highly elastic long-run residential electricity demand in California. Admittedly, two of the elasticities suggest a larger price-spillover effect. Baseline renewable electricity supply is likely inelastic (ε_{yb} small) and new fossil supply is likely inelastic (ε_{fa} small) (Aldy et al., 2023).¹⁵ However, the new fossil share θ_{fa} is negligible, so this latter factor does not hold much weight. Perhaps most importantly, baseline fossil supply has a large market share (θ_{fb} is large) and is relatively elastic (ε_{fb} is large), which would limit the price-driven spillover effects on new renewables in non-Energy Communities.

Overall, although infrastructure externalities could have positive or negative spillover effects, the empirical evidence suggests the net effect on nearby areas is a crowd-in in the short run and null in the long run. And, although the traditional price spillover effect would cause my analysis to over-estimate the subsidy effect, it seems likely this effect is small.

3.5 Long-run Effects

Combining the results of the previous sections, I calculate a long-run effect of the Energy Community Bonus on county-level capacity additions. The long-run effect accounts for the endogenous location choice of projects to enter the queue in an Energy Community, as well as the probability these projects accept their interconnection cost offers.

For some county j , the long-run effect of the bonus on capacity additions A_j

$$\tau_{LR} = E[A_j(D) \mid D = 1] - E[A_j(D) \mid D = 0], \quad (5)$$

where D is an indicator for receiving the subsidy.

To leverage the empirical estimates of the previous analyses, I model long-run capacity additions as

$$A_j(D) = Q_j(D)(1 - p_{p1,i}(D))(1 - p_{nf,i}(D))^{N_{nf}}(1 - p_{postIA}),$$

¹⁵Anecdotally, it is challenging to source new fossil generators. See: <https://www.utilitydive.com/news/5-year-waits-and-rising-costs-how-demand-is-redefining-the-gas-turbine-mar/813385/>

where $Q_j(D)$ is queue entry in county j , $p_{p1,i}(D)$ is the probability that queue entrants i withdraw through the first study, $p_{nf,i}(D)$ is the probability queue entrants withdraw after a non-first study, and N_{nf} is the number of non-first studies a project faces, which I assume is unaffected by the subsidy. p_{postIA} is the probability a project withdraws after its interconnection agreement. I also assume this is also unaffected by the subsidy. I discuss the magnitude and implications of these parameters and assumptions more below.

For the central estimate, I calculate $Q_j(D)$, $p_{p1,i}(D)$, and $p_{nf,i}(D)$ using the predicted values from my estimates in the analyses above. Queue entry regressions are measured in MW per county per half-year, so I multiply by two for annual values. Queue entry $Q_j(D)$ is estimated at the technology-level. For withdrawal probabilities $p_{p1,i}(D)$ and $p_{nf,i}(D)$, I apply the aggregate technology estimates to all technologies to preserve statistical power. For confidence intervals, I bootstrap the calculation by resampling the data underlying each estimate with replacement and re-estimating the equations.

I calculate the number of non-first studies N^{nf} as the mean number of non-first studies for operating projects in the data. The central estimate is 1.8. This number may under- or over-estimate the true mean number of non-first studies that recent queue entrants will face. It is an under-estimate because projects that tend to have more studies will not yet be operating, and not be included in this mean. It is an over-estimate because the projects included in the mean are older, and their studies occurred before the recent interconnection queue process improvements. I take the mean as-is for the central estimate, but it is an uncertain parameter, so I resample it in the bootstrapping procedure.

Some projects accept their cost offers and sign an “interconnection agreement,” but still never become operational, usually due to permitting challenges or unexpected cost changes. I use the LBNL QueuedUp data, which has the most data available on interconnection agreement signings, to calculate p_{postIA} from historical data (Rand et al., 2025). For projects that sign an interconnection agreement in 2017 through July 2025, I calculate p_{postIA} as the share of withdrawn projects among the projects that are either withdrawn or operational. I drop the half of these projects that are still active and whose post-interconnection-agreement survival or withdrawal is censored. I bootstrap this mean to account for the uncertainty this introduces.

Assuming that p_{postIA} is unaffected by the subsidy is necessary because not enough time elapsed between the Energy Community Bonus enactment in 2023 and the 2025 repeal to directly estimate the subsidy effect on this survival rate: The median time from interconnection agreement to commercial operation for projects operating in 2025 was about 30 months (Rand et al., 2025). This is on top of the time it takes to complete interconnection cost studies. Assuming p_{postIA} is unaffected by the subsidy would, if anything, likely understate the magnitude of the Energy Community Bonus treatment effect.

	Solar	Wind	Battery+Other	Aggregate
MW/county/year	13.2	3.3	14.5	31.6
	(2.2, 26.8)	(-0.8, 9.2)	(3.8, 30.5)	(7.2, 65.2)
Percent increase	259.0	135.0	364.0	289.0
	(43.0, 527.0)	(-33.0, 373.0)	(96.0, 765.0)	(66.0, 596.0)

Table 4: Long-run effect of the Energy Community Bonus on annual capacity additions from Equation 5. Percent increase is relative to predicted counterfactual without the subsidy. Bootstrapped 95 percent confidence intervals in parentheses.

Table 4 shows the result. In the long run, the Energy Community Bonus increases capacity additions in Energy Communities by about 30 MW per county per year. The percent estimates are noisily estimated because I bootstrap a ratio $((A_j(1) - A_j(0))/A_j(0))$. However, they show a tripling of overall capacity additions in Energy Communities due to the Bonus. Most of this is driven by solar and battery storage, and to a lesser extent, wind. The Bonus increases aggregate queue entry from 86 MW per county per year to 164 MW per county per year, a 92 percent increase. It increases the predicted survival probability from 13 percent to 26 percent. The combined effect is 32 MW per county per year, or a 289 percent increase from the baseline.

Moreover, the project survival probabilities are estimated from SPP and MISO data, which see the largest queue entry and capacity addition effects (see Appendix B). I am in the process of acquiring interconnection cost and survival data from PJM, which sees null queue entry and capacity addition effects. Including these data could lower the probability effects to better represent national average effects, and ultimately lower the estimated average long-run effects.

With these results in hand, the remainder of the paper combines the results with estimates of the magnitude of externalities caused by clean electricity generators to assess the marginal welfare effects of the Energy Community policy and clean electricity subsidies generally.

4 Emissions and Welfare Effects

I use two approaches to measure the welfare effects of clean electricity subsidies. The quantity changes measured in the previous section are the combined result of supply and demand changes.

Based on the simple intuition of Harberger triangles, the first approach uses this quantity change as a sufficient statistic for the welfare changes induced by clean electricity subsidies (Chetty, 2009). It is also useful for some back-of-the-envelope calculations on emissions effects. The second approach uses the MVPF (Hendren and Sprung-Keyser, 2020; Hahn et al., 2026).

I use the models to estimate the marginal effects of clean electricity subsidies in the US on aggregate. I then assess how the Energy Community Bonus compares to other policies in terms of emissions abated, welfare effects, and the MVPF. But first, I offer some general intuition about the welfare effects of place-based clean electricity subsidies.

4.1 The Welfare Effects of Place-Based Clean Electricity Subsidies

In the case of a carbon tax or a subsidy per carbon abated, it is clear that a uniform policy is more cost-effective than a place-based policy. Indeed, the whole point of a carbon pricing is to equalize abatement costs, and spatially differentiating the instrument undermines this. In the case of clean electricity subsidies, it is not obvious whether the place-based subsidy will improve or worsen the welfare relative to some counterfactual subsidy (e.g., a capacity-neutral or fiscally neutral subsidy). The reason is that clean electricity is an imperfect proxy for emissions abatement. The ITC is a subsidy per MW and the PTC a subsidy per MWh, but the externality per MW or MWh spatially varies throughout the country. Therefore, an abatement-cost minimizing subsidy also spatially varies.

There are two sources of spatial variation that drive the imperfection. The first is spatial variation in marginal emissions. Marginal emissions are the emissions that are generated by the marginal generator on the electricity grid (usually the most expensive generator), and that are displaced from an additional unit of lower-marginal-cost electricity, such as renewables. Clearly, generators do not internalize the social value of emissions abatement, so pricing proportionately to marginal emissions can improve outcomes.

Perhaps less clearly, the other source of imperfect pricing is geography, in particular windiness and sunniness. This is perhaps less clear because generators do, at least to some extent, internalize this value. Consider second-best optimal clean electricity subsidies in a case where marginal emissions are spatially uniform.¹⁶ The second-best optimal ITC spatially varies proportionately to sunniness and windiness, so a uniform instrument is surely suboptimal. But the second-best optimal PTC is uniform. Nonetheless, a marginal increase in ITC or PTC levels in sunnier or windier areas still improves welfare compared to a marginal increase in a less sunny or less windy area. The reason is

¹⁶“Second-best optimal” here refers to a policy that can only subsidize clean electricity in an economy where there is no other direct or indirect carbon pricing.

that the baseline PTC is below the second-best optimal level. So a marginal increase to the PTC in a windy area improves welfare more than a marginal increase in a non-windy area. Appendix F.1 shows this using a simple model.

With this intuition, whether the Energy Community Bonus improves or worsens welfare relative to counterfactual subsidies depends on the correlation between Energy Community designation, marginal emissions, and geography. I next describe a framework to quantify the welfare effects of clean electricity subsidies using the empirical results in this paper.

4.2 Model

4.2.1 Sufficient Statistics for Welfare

Denote x_j as a county-technology combination of a clean electricity capacity addition. For example, one unit of x_j may be one MW of solar capacity in some county. Let ϕ_j be the net present value (NPV) of an externality caused by one unit of x_j .

Consider a marginal change moving from one tax vector $\tau = (t_1, \dots, t_J)$ to another tax vector $\tau' = (t'_1, \dots, t'_J)$. The change in deadweight loss is

$$\Delta DWL = \frac{1}{2} \sum_j \sum_k (\phi_j - t_j + \phi_j - t'_j) \frac{dx_j}{dt_k} (t'_k - t_k). \quad (6)$$

The formal proof is in Appendix F.3. The intuition is from the Harberger triangle (Harberger, 1964). This formula is for the trapezoid that is formed as the difference between two deadweight loss (DWL) triangles. $\phi_j - t_j$ and $\phi_j - t'_j$ are the heights of the two magnitudes of mispricing caused by t_j and t'_j . $\frac{dx_j}{dt_k} (t'_k - t_k)$ is the quantity change in x_j caused by a change to the tax on x_k , the width of the trapezoid. The sum is over every county-technology combination, where the terms for $j \neq k$ are spillover effects. Similarly, the change in emissions caused by the policy change is

$$\Delta Emissions = \sum_j \sum_k e_j \frac{dx_j}{dt_k} (t'_k - t_k)$$

where e_j is measured in NPV since the change in emissions caused by the new clean electricity capacity is a stream of emissions changes into the future.

Importantly, the quantity change partial $\frac{dx_j}{dt_k}$ is not a demand or supply elasticity per se. It is the quantity change that results due to the net of supply and demand. This is the key benefit of writing the expression for the change in DWL this way. The quantity changes from the empirical analysis are sufficient statistics for welfare effects.

Deriving Equation 6 makes assumptions which have implications for the types of counterfactuals I can model. The first assumption is that the externalities ϕ_j are constant over the support of the change in x_j for all j . Thus, the model assumes the marginal generator is unchanged by the presence or absence of the Energy Community Bonus or any of the counterfactual policies.

The second assumption is that $\frac{\partial x_j}{\partial t_k}$ are constant. A consequence of this linearity assumption is that it implies that an increase in the subsidy rate linearly increases clean electricity, while convexly increasing fiscal costs.¹⁷

This undermines using a uniform, revenue-neutral subsidy counterfactual when evaluating the place-based subsidy. The revenue-neutral subsidy will dominate almost any place-based policy, barring extreme externality differences. Spreading the fiscal cost over more locations will cause more subsidy effect ($\frac{dx_j}{dt_j}$) per fiscal dollar because there is less fiscal cost penalty from the convexly increasing fiscal costs in any particular t_j . That means more renewable electricity and externality per dollar under any subsidy that covers more locations.

For a counterfactual comparison that is not driven by the linearity assumption, instead of comparing the Energy Community Bonus to a revenue-neutral subsidy, my preferred counterfactual benchmark is a capacity-neutral subsidy. This is a uniform subsidy that results in an equal amount of capacity as the Energy Community Bonus. The capacity-neutral subsidy is calculated as

$$\tilde{t} = \frac{\sum_j \frac{dx_j}{dt_j} EC_j t_j^{EC}}{\sum_j \frac{dx_j}{dt_j}},$$

where t_j^{EC} is the Energy Community bonus value for j (not including the baseline ITC or PTC values) and EC_j is equal to 1 if county j is an Energy Community. Note, this is not only the capacity-neutral uniform subsidy, but also the quantity-change-weighted mean value of the Energy Community Bonus across all counties and technologies. I will show below using data that the subsidy value depends on technology types. I therefore calculate these values, and use as counterfactuals the within-technology capacity-neutral subsidies.

4.2.2 Marginal Effects Model

The MVPF is a benefit-cost ratio,

$$MVPF = \frac{WTP}{Net\ Govt\ Cost} = \frac{x \cdot d\tau + \phi \cdot dx}{x \cdot d\tau + \tau \cdot dx} \quad (7)$$

¹⁷Fiscal costs are expressed as $\sum_j \sum_k t_j \frac{dx_j}{dt_k} t_k$. Assuming no spillovers, it is easy to see the convex functional form $\sum_j \frac{dx_j}{dt_k} t_j^2$. In terms of the Harberger triangle, this is the revenue rectangle increasing in both height and width with a subsidy increase.

where the numerator is societal willingness to pay (WTP) for a policy and the denominator is the net government cost. x , $d\tau$, ϕ , dx , and τ are J -length vectors, and the dot is the dot product. The components are the same as those in Equation 6, though this adds the requirement of knowing the size of the inframarginal capacity additions x_j .

Societal WTP for a marginal transfer is the magnitude of the transfer $x \cdot d\tau$ to inframarginal recipients, plus the externality created by the transfer $\phi \cdot dx$, where dx is the behavioral change caused by the policy. The marginal transfer recipients do not value the transfer because of the market distortion induced; they are exactly indifferent between receiving the transfer and the marginal unit of consumption (the envelope theorem). The fiscal cost in the denominator is composed of the same marginal transfer to the inframarginal recipients, plus the total value of the transfer to the marginal recipients.

The MVPF has the desirable property that a budget constrained social planner choosing between two policies would raise the most welfare by choosing the policy with the highest (welfare-weighted) MVPF. A policy with an MVPF greater than one raises social welfare, and a policy with an MVPF less than one lowers welfare. A Pigouvian tax has an MVPF equal to one.

Below, I consider an MVPF for raising the subsidy level by \$1 for all counties and all technologies,

$$MVPF = \frac{\sum_j x_j + \phi_j \sum_k \frac{\partial x_j}{\partial t_k}}{\sum_j x_j + t_j \sum_k \frac{\partial x_j}{\partial t_k}}. \quad (8)$$

I also consider another metric of interest to benchmark clean electricity subsidies against other climate policies, the marginal fiscal cost per ton of CO₂ abated from raising the subsidy level by one dollar:

$$\frac{\$Subsidy}{Tons} = \frac{\sum_j x_j + t_j \sum_k \frac{\partial x_j}{\partial t_k}}{\sum_j \sum_k \frac{\partial x_j}{\partial t_k} e_j}, \quad (9)$$

where e_j is again the NPV of the emissions abated by an additional MW of x_j .

Like the welfare calculation in Equation 6, calculating the MVPF in Equation 8 does not require an estimate of supply or demand elasticities. All that is needed is externality values, subsidy values, and estimates of the marginal effect of a subsidy change on capacity additions and the inframarginal capacity additions. The next section describes how I measure each of these components.

4.3 Data and Calculations for Emissions and Tax Credit Values

As alluded to in Section 2, I use forecast data of future electricity grid emissions, electricity prices, and capacity factors to calculate clean electricity externalities and tax credit values. I describe the process and calculation results in this section, with more detail available in Appendix F.2.

Emissions Abated by Clean Electricity (ϕ) Calculating the NPV of the emissions abated by one MW of technology-location-specific electricity capacity has three key ingredients: (1) the future stream of electricity generated by the MW of capacity (converting MW to MWh), (2) the emissions abated by that MWh (converting MWh to emissions), and (3) the value of those emissions. I calculate this stream of electricity and emissions at the county-by-hour level for every county in the US and every hour from 2025 to 2050.

For electricity generated by a MW, or capacity factors, I use 15 years of historical sunniness and windiness data to obtain the electricity generated by the optimal generator choice for each location based on data from the National Renewable Energy Laboratory (NREL) ReEDS model (Sergi et al., 2024). I use the Cambium 2024 model for a future stream of electricity prices, assuming batteries charge during the least expensive four hours and discharge during the most expensive four hours per day (Gagnon et al., 2024). To convert the generation to emissions abated, I use the Cambium model's marginal emissions projections.

For emissions results, I convert the stream of future emissions abatement to a present discounted value (PDV) assuming a 2 percent social discount rate on emissions abated. For results that value the emissions externality, I first multiply the Social Cost of Carbon (SCC) from EPA (2023) by the stream of abatement, and then discount the stream of valued abatement to a PDV.

Tax Credit Values (τ) I calculate the value of the baseline ITC and PTC and the Energy Community Bonus adder for one MW of solar, wind, or batteries in each county. Projects can choose which of the tax credits to claim. I calculate both tax credit values and assume the project takes whichever is most valuable per MW.

To calculate the ITC value, I use the 2024 Annual Technology Baseline model, which has the capital expenditures required per MW of solar, wind, or battery storage (NLR, 2024). The model chooses the solar panel and wind turbine that it estimates is most profitable at each location. The ITC baseline is 30 percent of this value, and the Energy Community Bonus an additional 10 percent.

Calculating the PTC value per MW is more complicated because it requires converting a 10-year subsidy per MWh to a subsidy per MW in PDV. I assume that project developers use a private discount rate of 8 percent when valuing the 10-year stream of values. For the purpose of weighing the PTC values against the externality values, I use a social discount rate of 2 percent. I use the same capacity factors as I do for marginal emissions to obtain 10 years of electricity generation per MW for every technology and every location. The PTC is a constant 2.75 cents per kWh in 2023, indexed to inflation, which I assume to be 2.5 percent per year based on the 2025 30-year inflation expectations

(FRED, 2025).

I use a more inclusive definition of Energy Communities for these welfare and emissions analyses than I do for the empirical analyses. Here I consider Energy Communities to be counties that meet the fossil fuel employment definition in all years, have at least one coal census tract, or at least 10 brownfields. Under this definition, 1,499 counties are Energy Communities, which is 48 percent of the counties in the US.

Appendix Figure 23 shows the result of the calculations. Wind takes the PTC and battery the ITC with rare exception. Solar takes the PTC in the sunniest parts of the country, and otherwise takes the ITC. The Energy Community Bonus is a larger adder to the ITC (10 percentage points) than the PTC (10 percent), so it makes the ITC the more valuable credit regardless of location for solar. It also nudges wind to claim the ITC in a few locations. The Energy Community Bonus value ranges from \$57 thousand to \$215 thousand. It is most valuable for locations and technologies that claim the ITC. The average private PDV of the Bonus per MW for batteries is \$171 thousand, for solar is \$146 thousand, and for wind is \$77 thousand.

Inframarginal Capacity Additions x_j and Marginal Policy Effects ($\frac{\partial x_j}{\partial t_k}$) Based on the empirics and argument in Section 3.4, I assume zero spillover effects such that $\frac{\partial x_j}{\partial t_k} = 0$ for all $j \neq k$. This greatly simplifies the welfare Equation 6 and the MVPF Equation 8. I use the aggregate long-run treatment effects (i.e., from Table 4) as the national average treatment effect.

I incorporate location-by-technology heterogeneity for the average aggregate estimate based on the historical interconnection queue data in each area. The resulting average share of each technology is 43 percent solar, 17 percent wind, and 40 percent battery with sensible regional variation. The long-run aggregate treatment effect in Table 4 is about 42 percent solar, 11 percent wind, and 46 percent battery and other. The consistency of these values supports the approach assuming a uniform treatment effect varying according to technology prevalence.

I calculate $\partial x_j / \partial t_j = \hat{\beta}_j / t_j^{EC}$, where $\hat{\beta}_j$ is the aggregate treatment effect multiplied by its county j 's share of that technology and t_j^{EC} is the county-technology-specific Energy Community Bonus value not including the baseline tax credit. The inframarginal capacities x_j are the predicted counterfactual treatment effects.

4.4 Results

Table 5 presents the national MVPF estimates of Equation 8 and marginal fiscal cost per ton Equation 9. The MVPF is highest for solar and wind, and lower for batteries. This difference is

	Solar	Wind	Battery+Other	Aggregate
MVPF	8.56 (6.11, 9.11)	7.36 (6.62, 7.48)	2.60 (2.06, 2.73)	6.49 (5.12, 6.77)
Marginal subsidy per ton	29.07 (27.07, 42.40)	33.24 (32.60, 37.49)	105.56 (97.73, 157.54)	38.37 (36.49, 50.84)

Table 5: MVPFs and marginal fiscal cost per ton as described in Equations 8 and 9 and 95 percent confidence intervals. Confidence intervals generated using the bootstrapped estimates of the policy effect from the aggregate effect estimates.

driven by the fact that batteries need to charge, so they both emit and abate. However, the MVPF for all technologies is quite high. For example, over \$7 of social benefit is created for every \$1 of fiscal transfer toward a wind subsidy. This MVPF is higher than the equivalent MVPF in Hahn et al. (2026), about 3.6, so using an updated policy elasticity appreciably raises the MVPF.¹⁸ Marginal fiscal costs per ton of CO₂ provide the same policy ordering as the MVPF. By raising all the subsidy levels by \$1, the resultant fiscal transfer per ton of CO₂ is \$38.

4.4.1 A Trade-off or Win-Win for Efficiency and Emissions?

Table 6 presents the results calculating Harberger triangles and emissions abated by the Energy Community Bonus compared to a nationwide uniform subsidy that results in an equal amount of new capacity of solar, wind, and battery technology. The Energy Community Bonus improves the emissions and welfare effects of solar and wind subsidies. It worsens the effects for battery subsidies. For every year the Energy Community Bonus is in place for solar and wind, it results in an additional \$800 million of NPV of annual social surplus. However, the equivalent loss is \$1.1 billion per year for batteries. The magnitude of the battery welfare loss is higher than the relative magnitude of its emissions. This is because the welfare calculation weights the effect not only by the SCC, but also by the deviation from the social cost, including the baseline tax credit. The Energy Community Bonus causes a larger shift away from the social cost for batteries than it does for solar and wind combined.

Figure 8 assesses the question in two other ways, with similar qualitative results. First, I plot the national MVPF of a \$1 subsidy increase, and the MVPF of an Energy-Community-only \$1 subsidy

¹⁸The 3.6 MVPF is based on their Figure 2.A, counting only the global environmental externality value (4.931/1.354).

	ECB	Capacity Neutral	Uniform
	Value	Value	Difference
Emissions abated (Mt CO ₂ e)			
Solar	388.3	383.1	5.1
Wind	207.8	202.0	5.8
Battery	99.1	102.2	-3.1
Welfare from abatement (\$B)			
Solar	41.4	41.3	0.1
Wind	21.9	21.2	0.7
Battery	6.7	7.8	-1.1

Table 6: Welfare effects of the Energy Community Bonus compared to a capacity-neutral uniform subsidy. Values are in NPV of emissions or welfare caused by one year of long-run capacity effect.

increase. This compares the MVPF of a marginal place-based subsidy to a marginal national subsidy. To get a sense of magnitude, the histogram shows the MVPF of a \$1 subsidy increase to 1,499 random counties (the same number as there are Energy Communities), iterated 10,000 times. The Energy Community Bonus is far better than random for solar, wind, and the aggregate subsidy. For batteries, its within the tails of the random histogram, but still substantially worse than random.

Regardless of how this question is assessed, the qualitative takeaway about the efficiency of the Energy Community Bonus is the same. Better for solar and wind, worse for batteries. However, even the worst randomly targeted clean electricity subsidies have relatively high MVPFs.

5 Conclusion

A marginal subsidy to utility-scale clean electricity has large, positive effects on new electricity capacity in both the short and long run. Despite skepticism that the subsidies are inframarginal, or that NIMBY-ism is enough to limit subsidy effects, it seems that there is substantial elasticity in this market. The marginal welfare and emissions effects are large both because the policy effects are large and positive and because the wedge between the social benefit and subsidy levels is large.

I conclude with two caveats on the interpretation of this analysis. First, the welfare effects here only include the global CO₂ externality. There are other externalities generated by new generators, both positive and negative, such as local labor market distortions, local disamenities, learning-by-

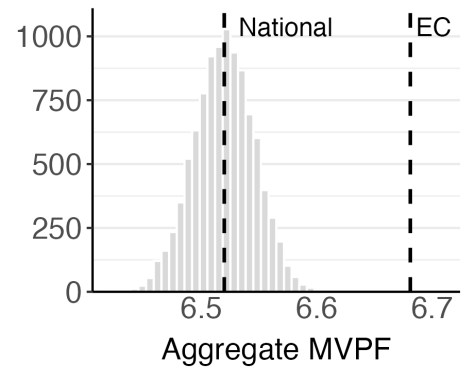
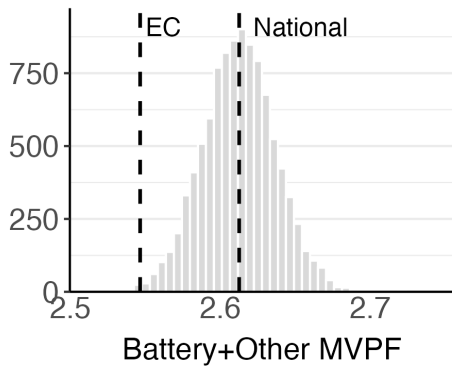
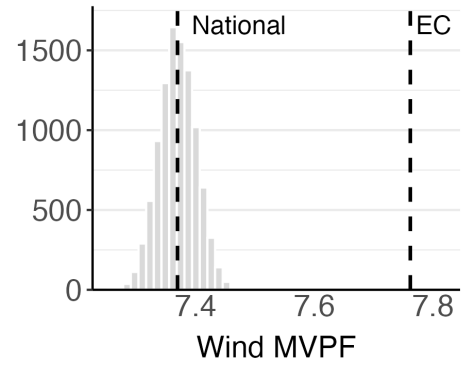
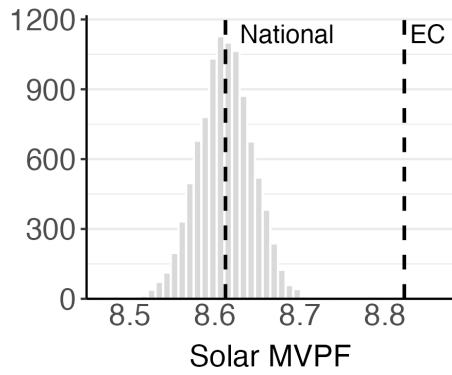


Figure 8: The MVPF of an additional dollar to Energy Communities versus a random choice of counties.

doing, and air pollution. Second, the clean electricity industry changes regularly, which is largely what motivated this analysis. The estimated subsidy effects are internally valid for the years in the analysis, but not necessarily static primitives.

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A Data Cleaning

A.1 Interconnection Queue Data Cleaning and Missing Data Handling

The queue data include observations from all contiguous 48 states. I drop observations in Mexico and Canada from the estimating sample. I also filter out load and transmission projects that are in queues because these are not eligible for the ECB. That leaves 37,535 observations of projects from 1995 to August 2025. I need all queued projects to have location data precise enough to join them to county-level Energy Community data. Most projects list FIPS codes that are sufficient for this join. 2 projects have no FIPS codes, but include coordinate data that I use to join the projects to counties. 12 projects are associated with cities, not counties, so I manually join them to their containing county. 563 observations have no data on FIPS codes, coordinates, county names, or city names. I drop these observations, and am left with 36,972 projects. **NEED TO REVISIT THESE PROJECT-LEVEL NUMBERS**

The data lists multiple FIPS codes for 336 of projects. However, it is important to connect projects to one county, as only one county can determine its EC status.¹⁹ This adds an additional dimension of uncertainty regarding whether these projects are indeed EC projects. For project-level analyses, I split multi-county projects into multiple observations, and weight these observations by the inverse of the number of counties associated with the project. For example, a project that lists two counties is associated with both counties, but only included in regressions as a half-project. For county-level analyses, I again split multi-county projects into multiple observations. I count fractional projects for to each county, and evenly divide the MW of the multi-county project across the listed counties.

Another issue that needs to be addressed with the Interconnection.FYI dataset is missing data. Interconnection.FYI collects data by scraping balancing authority websites and, when necessary, directly contacting balancing authorities. The result is coverage of grid operators that account for about 97 percent of electricity capacity. However, that leaves about 3 percent of US electricity generation missing from the data (Rand et al., 2025). For the county-level queue entry analysis, I identify and drop counties that are covered by the grid operators that are not sampled by the Interconnection.FYI data. Without this correction, the count of projects in MW in these counties would be biased downward. I use data from the US Department of Energy (DOE) with the approximate spatial coverage area of balancing authorities in the US to identify and drop counties that are covered by unsampled

¹⁹A project is considered an EC project if 50 percent or more of its nameplate capacity or its footprint is in an EC. For offshore generation, this is determined by where the equipment connects to land (Service, 2025; Treasury, 2023a,e).

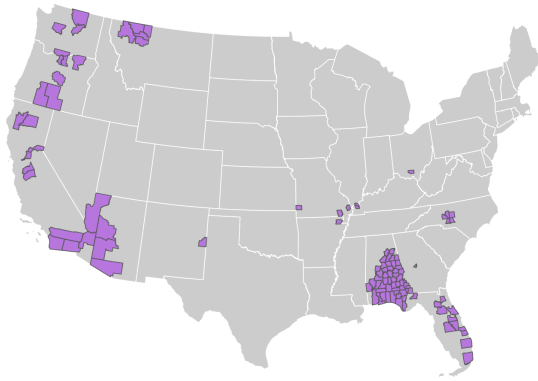


Figure 9: Counties covered by balancing authorities that are not in the data sample.

balancing authorities.²⁰ Specifically, I find balancing authorities in the DOE data that are not in the interconnection queue data or listed in the Interconnection.FYI metadata on grid operators covered. This drop 107 counties, shown in Figure 9.

A second type of missing information in the Interconnection.FYI is somewhat more consequential. All of the sampled balancing authorities include information on project status (whether it is active, withdrawn, or operational), but some do not include the dates of status changes. These dates are crucial for the analyses in this paper, as identification relies on the timing of ECB policy enactment and the timing of status changes. For queue entry analyses, I drop counties covered by ERCOT, Georgia Transmission, Dominion Energy South Carolina, or Idaho Power Company. ERCOT in particular contains almost 20 percent of the active capacity in US queues as year-end 2025. Although the ERCOT website updates is queue data monthly, this is apparently not a reliable way to track new queue entrants. For example, some projects are recorded as signing an interconnection agreement in August 2024, but do not appear in ERCOT’s queue reports until August, September, or October 2024.

For dropping counties, I assess whether a county is managed by a certain grid operator in two ways, and drop the counties if it is identified by either method. First, I consider counties with a queued project managed a balancing authority as covered by that balancing authority. Second, I use the DOE spatial data and select counties that have more than 1 percent of their area covered by a balancing authority with a lot of missing data. The procedure drops ultimately drops 586 counties for the queue entry analyses. After this procedure, 0.6 percent of the remaining projects have a null

²⁰<https://atlas.eia.gov/datasets/balancing-authorities/explore>

queue entry date. These are dropped from the county-level queue entry analyses. The final estimating sample for county-level queue entry analyses includes **28,239** projects and 2,522 counties.

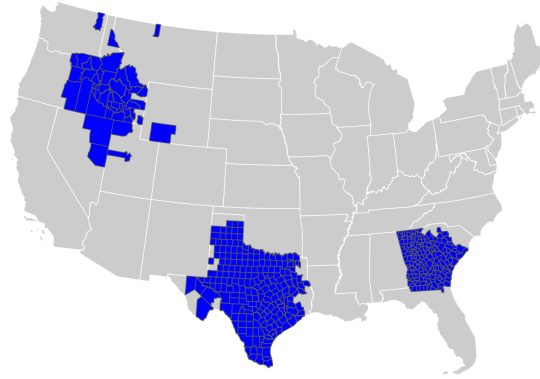


Figure 10: High proportion null queue entry dates.

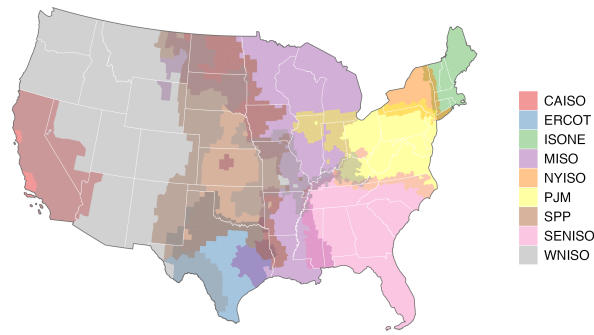


Figure 11: Counties managed by each grid operator.

- Batteries are not assuming 100% roundtrip efficiency. That's captured in the MEs.
- Equally distributing capacity for hybrid projects for the probability regressions, same as queue regressions
- Not disaggregating projects for costs, though, because less clear how to split the costs
- Costs inflation adjusted to 2024 using GDP price deflator <https://fred.stlouisfed.org/series/A191RD3A086NBE>

A.2 Energy Community Data Cleaning

Treasury (2023a,b,c,d,e) define the Energy Community definition data. The list of NAICS codes used to define fossil fuel employment as it pertains to Energy Communities was expanded in March 2024, over a year after the Bonus had been made available. I use the initial fossil fuel employment designation for defining Energy Communities in 2023 and earlier, and the updated definition in 2024 and later.

Related, it is somewhat ambiguous which is the correct County Business Patterns (CBP) data to use. CBP for a given year is usually released with about a two year lag. In 2023, it is clear that the official EC definitions used the 2021 CBP, as that was the most recent CBP available. When the 2024 official EC dataset was released, the 2021 CBP was still the most recent version of the CBP, so it must also consider the 2021 CBP. The 2022 CBP was released a few months later. However, the 2022 CBP would have been available for tax year 2024. I use the most recent CBP data that was available in the tax-year. It turns out this does not matter because no MSA or NMSAs newly pass the 0.17 percent fossil fuel employment threshold in 2024 or 2025. But it is perhaps notable for future work that attempts to identify Energy Communities.

To identify EC Brownfields in the EPA Facility Registry service, I follow the Treasury Guidances, as well as the EnergyCommunities.gov Frequently Asked Questions page from 2024, which included a question comparing EPA brownfields to Energy Community Brownfields. I drop observations that are Superfund sites and observations associated with the Resource and Conservation Recovery Act, both of which are not eligible Energy Community brownfields. Parcels that only have pollution due to petroleum contamination are also ineligible, so I drop observations that are likely a gas station or petroleum processing facility. I drop facilities with primary names that include the name of about a dozen of the largest gas station brands in the US, or include the words petroleum, gasoline, or gas station (e.g., Mobil, BP, etc.).

Even if this process perfectly identifies the subset of EPA Brownfields that are also EC Brownfields, it will still miss locations that may be eligible as Energy Community Brownfields and have not yet been identified by an IRS environmental assessment. However, given that Brownfields tend to be spatially clustered, this measure should give a good proxy for Brownfields even if it is imperfect.

B Treatment Heterogeneity by Grid Operator

I re-run the SDID at the grid operator level, accordingly dropping the operator-by-year fixed effects. Many ISOs have an insufficient number of treatments or controls, so I group them based on geographic proximity. Unsurprisingly, running estimating treatment effects for these sparse grid operators alone produces noisy and null results. I group PJM, ISO-NE, and NYISO because ISO-NE and NYISO combined only have 8 treated Energy Community counties, 4 neighboring counties to investigate spatial spillovers, and about 32 controls. CAISO only has 3 eligible control counties, and ERCOT has 80 Energy Community counties compared to 20 controls, so I group them both with non-ISOs. Jackknife standard errors are in parentheses, following Arkhangelsky et al. (2021) Algorithm 3. Bootstrapping is no longer necessary because there is not first stage fixed effect estimation.

Results are in Table 7 for Energy Community direct effects, and Table 8 for neighboring county spillover effects. Each cell in this table is a separate SDID run. For legibility, I include asterisks for significance according to multiple hypothesis testing Bonferroni-adjusted p-values. I treat these as 32 tests, one for each cell within the tables.

For effects on Energy Communities, the positive US-level treatment effects seem to be driven by MISO, SPP, CAISO, ERCOT, and non-ISOs. There are no significant effects in the northeast ISO regions.

Neighbor spillover effects are noisier due to a smaller number of treated counties, especially SPP. Nothing is significant. A couple positive queue entry spillovers have their significance undermined by the multiple hypothesis testing correction, including solar queue entry in PJM, NYISO, and ISO-NE, and battery queue entry in MISO, and aggregate technology capacity additions in MISO, CAISO, ERCOT, and non-ISOs.

Overall, these results suggest heterogeneity, but also a lot of insignificance and limited statistical power. In assessing subsidy effects, I advocate for using the full US level sample because the most suitable control counties are not necessarily in the same ISO, and pooling the regions offers more statistical power.

		MISO	SPP	PJM, NYISO, ISO-NE	CAISO, ERCOT, non-ISO
Cap. Add.	Solar	10.8*** (2.6)	17.6 (8.2)	1.0 (2.1)	16.9*** (3.9)
	Wind	1.9 (1.3)	-1.4 (3.5)	-0.6 (0.6)	-0.7 (1.9)
	Battery	2.3 (1.3)	4.1 (2.8)	-0.2 (0.3)	13.5** (3.8)
	Aggregate	14.1*** (3.5)	21.2 (10.9)	-0.6 (2.4)	31.5*** (7.1)
	N treated / control	327 / 353	85 / 401	244 / 166	360 / 597
Queue	Solar	24.8*** (5.2)	23.1 (14.7)	-1.6 (4.1)	-1.2 (10.4)
	Wind	-5.5 (6.5)	15.3 (12.3)	-0.6 (2.2)	-13.1 (11.4)
	Battery	13.5*** (2.6)	23.0 (11.6)	-13.7 (7.0)	16.7 (11.4)
	Aggregate	19.1 (9.1)	56.6 (30.0)	-14.9 (10.4)	36.2 (23.3)
	N treated / control	292 / 351	49 / 388	243 / 166	242 / 386

Table 7: Energy Community treatment effect results, disaggregating grid operators. Capacity addition outcomes are in MW per county per year. Queue entry outcomes are in MW per county per half-year. Jackknife SEs in parentheses. Discrepancies between queue entry and operating regressions are also due to missing queue data or queue date data. Most notably, ERCOT drops out of the queue entry results.

C Robustness Checks for Queue Entry and Capacity Additions

		MISO	SPP	PJM, NYISO, ISO-NE	CAISO, ERCOT, non-ISO
Cap. Add.	Solar	7.2 (3.7)	4.6 (4.1)	7.0 (6.7)	3.6 (3.6)
	Wind	2.9 (2.6)	-3.6 (4.5)	-2.9 (2.3)	0.7 (2.4)
	Battery	-0.2 (0.6)	-0.5 (0.4)	-0.1 (0.6)	4.4 (2.2)
	Aggregate	9.5 (4.9)	0.8 (5.7)	5.0 (9.1)	10.6 (5.6)
	N treated / control	87 / 353	27 / 401	45 / 166	120 / 597
Queue	Solar	2.8 (8.2)	-8.0 (3.8)	3.5 (5.8)	0.3 (6.0)
	Wind	-2.0 (5.6)	-10.3 (14.3)	-9.7 (9.1)	-0.3 (4.0)
	Battery	6.7 (3.2)	-1.0 (5.3)	-5.7 (8.1)	0.0 (4.9)
	Aggregate	2.4 (10.1)	-30.0 (21.5)	-18.0 (17.4)	-2.7 (14.0)
	N treated / control	86 / 351	24 / 388	45 / 166	94 / 386

Table 8: Neighbor spillover effect results, disaggregating grid operators. Capacity addition outcomes are in MW per county per year. Queue entry outcomes are in MW per county per half-year. Jackknife SEs in parentheses. Discrepancies between queue entry and operating regressions are also due to missing queue data or queue date data. Most notably, ERCOT drops out of the queue entry results.

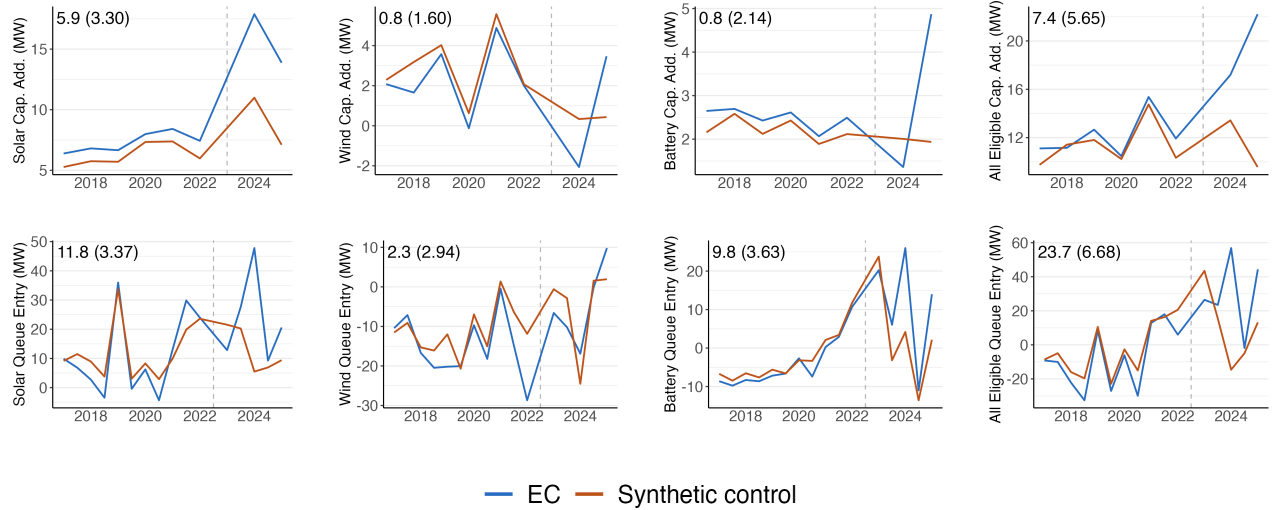


Figure 12: Capacity addition (top row) and queue entry (bottom row) estimating SDID Equation 2, dropping Arizona, California, Nevada, and New Mexico.

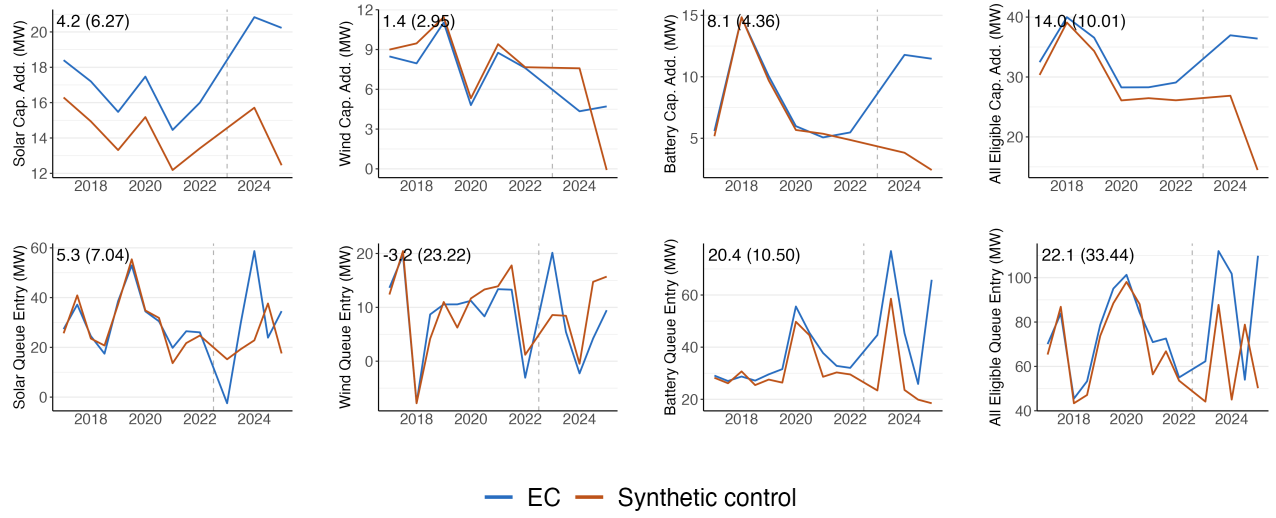


Figure 13: Capacity addition (top row) and queue entry (bottom row) estimating SDID Equation 2, with adding a state-by-year fixed effect in addition to the grid-operator-by-year fixed effect.

C.1 State-by-year Fixed Effects and Dropping California

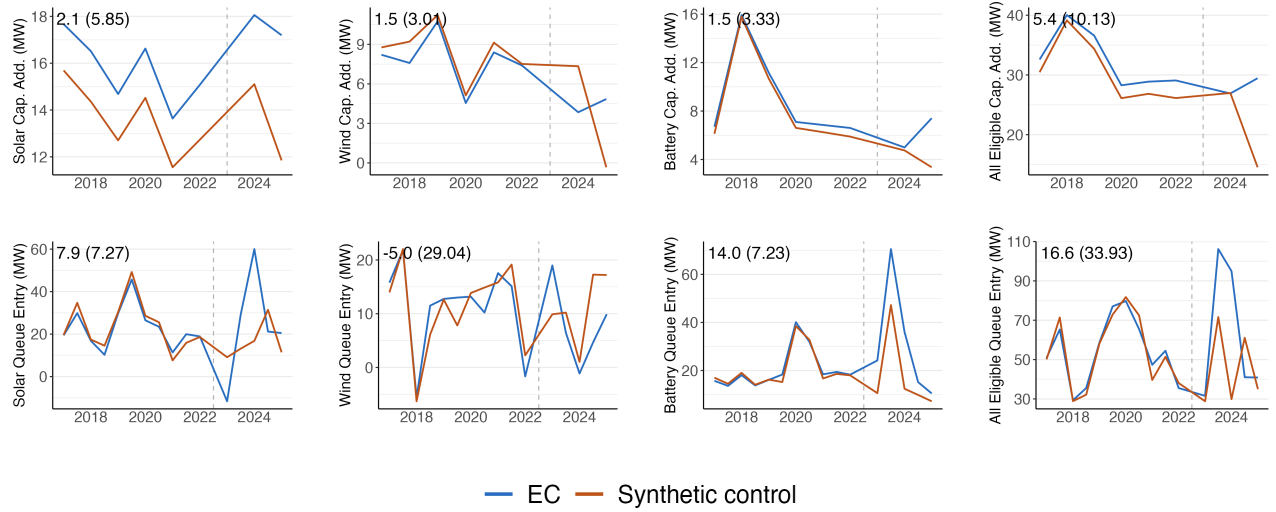


Figure 14: Capacity addition (top row) and queue entry (bottom row) estimating SDID Equation 2, with adding a state-by-year fixed effect in addition to the grid-operator-by-year fixed effect, and also dropping California.

D Supplementary Figures for Project-level Analyses

D.1 Interconnection Costs

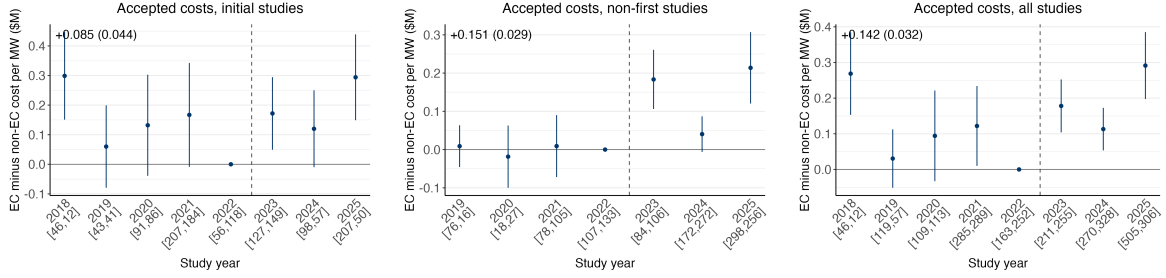


Figure 15: Event studies for data underlying Figure 6, separating initial studies and non-first studies. Event studies include study year and technology fixed effects. Capacity-weighted. Numbers in brackets are number of project observations in Energy Communities and non-Energy Communities, respectively. Standard errors clustered at the county level.

D.2 Probability of Queue Withdrawal

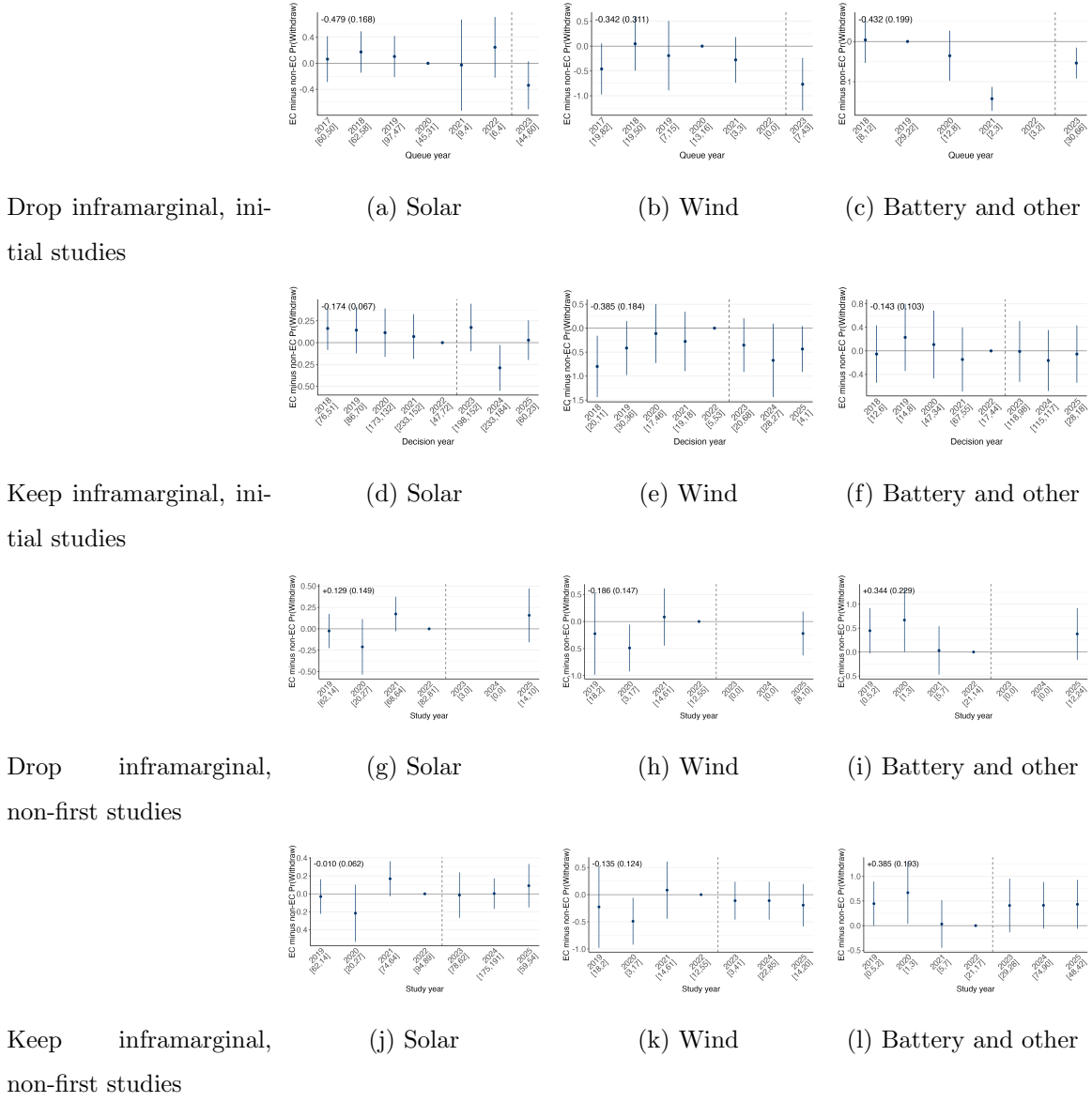


Figure 16: Figures 15 and ??, subsetting the sample on technology type. Pooled coefficient and standard errors reported at the top left. Standard errors are clustered at the county level. Studies for battery projects are rare before 2019, so I fix 2019 to be the reference year.

E Spillover Effects

E.1 Estimation with Treatment-Neighbors as the Treated Group

E.2 Equilibrium Price Spillover Model

Consider a stylized market for electricity in the long run. There is an aggregate demand for electricity that depends on price, $x(p)$. I assume demand is decreasing in price. Since this is a static, long-run model, the relevant price p is some aggregator of current and future electricity prices. Each county j supplies electricity from new renewable capacity additions $y_j^a(p)$, new fossil fuel additions $f_j^a(p)$, and baseline renewable and fossil fuel capacity $y_j^b(p)$ and $f_j^b(p)$. All supply functions are assumed increasing in price.

The market clears when aggregate demand for new capacity equals the sum of new capacity supplied by each county,

$$x(p) = \sum_j y_j^a(p + sEC_j) + y_j^b(p) + f_j^a(p) + f_j^b(p),$$

where t is a subsidy awarded if county j is an Energy Community, captured by the indicator EC_j .

Now consider a marginal change to subsidy s . Differentiating the market-clearing equation with respect to the subsidy yields the equilibrium change in price,

$$p' = \frac{\sum_j y_j^{a'} EC_j}{x' - \left(\sum_j y_j^{a'} + y_j^{b'} + f_j^{a'} + f_j^{b'} \right)} < 0,$$

where prime on price indicates its partial derivative with respect to the subsidy, and on supply and demand indicate their partial derivatives with respect to price. The inequality follows from assuming demand slopes down and supply slopes up. Intuitively, subsidizing supply lowers the equilibrium price.

Denote aggregate supply functions by dropping the j subscript, and denote quantities demanded and supplied as the same variables without parentheses (i.e., x total quantity demanded, y_j^a the total subsidy-eligible new electricity in county j). Then, we can write the equilibrium change in price in terms of elasticities and the share of total electricity that comes from new supply in Energy Communities,

$$\begin{aligned} p' &= \frac{\sum_j y_j^{a'} EC_j}{x' - y'} \\ &= \frac{\sum_j \frac{y_j^a}{p} \varepsilon_{y_j^a} EC_j}{\varepsilon_x \frac{x}{p} - \frac{y^a}{p} \varepsilon_{y^a} - \frac{f^a}{p} \varepsilon_{f^a} - \frac{y^b}{p} \varepsilon_{y^b} - \frac{f^b}{p} \varepsilon_{f^b}} \\ &= \frac{\sum_j \theta_{y_j^a} \varepsilon_{y_j^a} EC_j}{\varepsilon_x - \theta_{y^a} \varepsilon_{y^a} - \theta_{f^a} \varepsilon_{f^a} - \theta_{y^b} \varepsilon_{y^b} - \theta_{f^b} \varepsilon_{f^b}}. \end{aligned}$$

The first equality applies the definition of aggregate supply. The second equality substitutes the elasticity of demand $\varepsilon_x = x' \frac{p}{x}$, the elasticity of aggregate supply $\varepsilon_y = y' \frac{p}{y}$, and the county-level

subsidy-eligible elasticities of supply $\varepsilon_{y_j^a} = y_j^{a'} \frac{p}{y_j^a}$. The last equality comes from the fact that aggregate quantity supplied Y equals aggregate demand X in equilibrium, where $\theta_{y_j} = y_j/X$ is the share of *total* electricity (or total capacity in a capacity market) coming from new renewables in county j .

Finally, consider an estimate of subsidy treatment effects that comes from a subsidized county, $j = E$ to an unsubsidized county $j = N$. Since these areas are comparable, I assume the counties have equal supply functions for new electricity. Such a comparison would over-estimate the true subsidy effect by the multiplier

$$\begin{aligned} \frac{dy_E - dy_N}{dy_E} &= 1 - \frac{dy_N}{dy_E} \\ &= 1 - \frac{g_N^{a'} p'}{g_E^{a'} (p' + 1)} \\ &= 1 - \frac{p'}{(p' + 1)} \\ &= 1 - \frac{\sum_j \theta_{y_j^a} \varepsilon_{y_j} EC_j}{\varepsilon_x - \theta_{fa} \varepsilon_{fa} - \theta_{yb} \varepsilon_{yb} - \theta_{fb} \varepsilon_{fb} - \sum_j \theta_{y_j^a} \varepsilon_{y_j^a} (1 - EC_j)} \end{aligned}$$

The first and second lines are algebra and substituting $dg_N = g_N' p'$ and $dg_E = g_E' (p' + 1)$, the third comes from the assumption the two counties are comparable, and the last line substitutes in p' and re-arranges. θ_{fa} and ε_{fa} and the share of total electricity from and elasticity of new fossil fuel additions. The other variables are defined similarly.

Note that an alternative way to consider this model is with an inelastic baseline capacity. The key results are all the same. In particular, the numerator still includes $\sum_j \theta_{y_j^a} EC_j$, the share of *total, gridwide* electricity capacity that is from new additions in Energy Communities.

F Emissions and Welfare Model

F.1 Models for Intuition on Place-based Clean Electricity Subsidies

Consider an aggregate demand for electricity $x(p)$, measured in MWh , which yields societal utility $u(x(p))$, where $x' < 0$, $u' > 0$ and $u'' < 0$. The economy also features a numeraire x_0 that delivers linear utility with a price normalized to one.

J heterogeneous sources of electricity supply electricity capacity in perfect competition with supply functions $y_j(p)$, measured in MW , with $y'_j > 0$ for all j . Some sources are more productive than other areas (i.e., sunnier and windier), such that y_j capacity produces $a_j y_j$ units of electricity, where a_j is a capacity factor. Source-specific cost functions $c_j(y_j(p))$ have $c' > 0$ and $c'' > 0$.

Finally, electricity produces a source-specific externality ϕ_j . The market clears when $x(p) = \sum_j a_j y_j(p)$.

Baseline welfare is given by

$$W = U(x(p)) - px(p) + \sum_j pa_j y_j(p) - c_j(y_j(p)) + \phi_j a_j y_j(p).$$

With a social planner using a production subsidy, welfare is

$$W^p = U(x(p)) - px(p) + \sum_j (p + s_j^p) a_j y_j(p + s_j) - c_j(y_j(p + s_j^p)) + \phi_j a_j y_j(p + s_j^p) - s_j^p a_j y_j,$$

where s_j^p is a source-specific subsidy (or tax) per MWh. With an investment subsidy, welfare becomes

$$W^i = U(x(p)) - px(p) + \sum_j pa_j y_j(p) - c_j(y_j(p)) + s_j^i y_j + \phi_j a_j y_j(p) - s_j^i y_j,$$

where s_j^i is a source-specific subsidy (or tax) per MW. This clearly cancels out in the welfare function, but affects the result as it enters the producers' optimal capacity choice.

Differentiating the welfare function with respect to the subsidy for some source k yields first order conditions to obtain the optimal subsidies,

$$\begin{aligned} \frac{\partial W^p}{\partial s_k^p} &= (\phi_k - s_k^p) a_k \frac{\partial y_k}{\partial s_k^p} + \sum_{j \neq k} (\phi_j - s_j^p) a_j \frac{\partial y_j}{\partial s_k^p} = 0, \\ \frac{\partial W^i}{\partial s_k^i} &= (\phi_k a_k - s_k^i) \frac{\partial y_k}{\partial s_k^i} + \sum_{j \neq k} (\phi_j a_j - s_j^i) \frac{\partial y_j}{\partial s_k^i} = 0. \end{aligned}$$

The second-best optimal production subsidy is $s_k^{p*} = \phi_k$ and investment subsidy is $s_k^{i*} = \phi_k a_k$ for all k . That is, the optimal production subsidy does not depend on geography, while the optimal investment subsidy does.

In the case that subsidies are for some reason constrained to be sub-optimal, however, the change in welfare caused by a unit-increase in production subsidy still depends on geography a_k . In the case of under-subsidizing, $s_k^p < \phi_k$, a marginal subsidy increase to a more productive source (higher a_k) yields a higher social benefit. In the case of over-subsidizing, $s_k^p > \phi_k$, and a marginal subsidy increase to a more productive patch is more distortionary and implies greater social cost.

The point is, for investment subsidies, moving from a uniform to a place-based subsidy may improve or worsen welfare because the initial subsidy is obviously sub-optimal under any heterogeneity in externalities or geography. For production subsidies, spatially differentiating the subsidy by geography is not clearly welfare improving. However, in the case that the baseline uniform subsidy is too low, differentiating by geography still has ambiguous welfare effects compared to some other capacity-neutral or fiscally-neutral subsidy.

F.2 Emissions and Tax Data and Calculations

F.2.1 Emissions Abated by Clean Electricity

I rely on the National Lab of the Rockies (NLR) Cambium 2024 model to measure the emissions abated by a marginal clean electricity generator (Gagnon et al., 2024). In particular, I use their “Mid-Case” scenario, short-run marginal emissions projections. The short-run marginal emissions are the emissions that would occur from a marginal increase in a region’s load. I use Cambium’s busbar adjustment factor to convert to the marginal emissions that would arise from a marginal increase in the region’s generation. The only difference between these is the assumed transmission losses from generation or distribution. The data are at the “pzone” level, which are clusters of counties. There are 133 pzones. The marginal emissions data are zone-by-hour-of-year projections from 2025 to 2050 at five-year intervals. I linearly interpolate between the five year intervals at the pzone-by-hour-of-year level to create annual data.

I use Cambium’s short-run marginal emissions instead of their long-run marginal emissions. Their long-run marginal emissions product models a country-wide shock to electricity load and reports its effect on local emissions. This is a fundamentally different thought-experiment from that of the short-run marginal emissions rate, and the short-run marginal emission is what matches the nationwide emissions effect due to a marginal generation increase.

I choose to use a modeled marginal emission, instead of the many empirical estimates available the literature (e.g., Siler-Evans et al. (2012); Graff-Zivin et al. (2014); Holland et al. (2022, 2024)). While there is a strong justification to use the empirical estimates for present-day empirically estimated

marginal emissions, the marginal emission is likely to change in the future. I assume new capacity built in 2025 is online for 26 years until 2050. Although marginal emissions are stable relative to average emissions, the forecast is better suited than the empirical estimates for such a long time horizon (Holland et al., 2022).

My outcome variables are measured in MW of nameplate capacity, but it is MWh of electricity generation that abates electricity. I convert MW to MWh using county-level capacity factors for solar and wind technologies. This depends on location and the type of wind turbine or solar panel used. The 2024 County-Level Hourly Renewable Capacity Factor Dataset, which underlies NREL’s ReEDS model (Sergi et al., 2024). The wind data county-by-hour capacity factors for 15 weather years, 2007 to 2013 and 2016 to 2023. There are values for multiple different “resource classes” in each county. I assume that the wind generation takes the capacity factor of the best resource class in the county. I average this value across the 15 weather years for the final measure of each county’s hour-of-year wind capacity factor. I use a similar procedure to obtain solar capacity factors.

For battery capacity factors, I assume all batteries added to the grid charge and discharge 4 MWh per MW per day. Cambium provides projections for wholesale electricity prices. I interpolate to obtain annual prices in the same way as I do marginal emissions. I assume that batteries charge when it is least expensive, emitting the marginal emission, and discharge when electricity is most expensive, abating the marginal emission. I use random tie-breakers in the relatively rare instance that there are many hours in the day with the same electricity price.

Finally, I convert all the emissions abated and emitted to present discounted value (PDV) by county-technology, again assuming the capacity survives for 26 years beginning in 2025 and ending in 2050. I assume a 2 percent social discount rate on all future emissions. For emissions outcomes, this is the final metric.

To value the emissions, I use the social cost of carbon (SCC) from 2 percent discount rate scenario of EPA (2023). This value increases into the future, and I value the emissions abated at future dates at its future SCC level before convertintg the monetized emissions to a PDV. The outcomes are shown in Figure ??.

F.2.2 Tax Credit Values and Policy Changes

For every technology, I calculate the value of the PTC per MW of capacity and the value of the ITC per MW of capacity in every county. I assume the maximum of those two values is the value of the tax credit per MW by technology and locations.

The PTC per MW requires calculating how much electricity a MW of each technology in each location would generate for its first 10 years online. I use the same capacity factors from the previous section to obtain this stream of future electricity generation. I assume all projects meet the wage and apprenticeship requirements, such that the value of the PTC in 2023 is 2.75 cents per kWh. The tax credit is inflation-adjusted. I increase the tax credit at 2.5 percent per year, based on the 30-year expected inflation in 2025 (FRED, 2025). Lastly, I must discount the tax credit to a PDV such that its comparable to the ITC, which accrues immediately. I assume developers use a private discount rate of 8 percent to value the subsidy.

The ITC calculation is more straightforward. I obtain the cost per MW to build a new wind, solar, or battery generator from National Lab of the Rockies' Annual Technology Baseline (NLR, 2024). I use the capital expenditure value, which includes "overnight costs" as well as grid construction costs. The capital expenditures depend on the type of generator built (e.g., wind turbine size or solar panel capacity factor), which on a location's resource class. ATB models the profitability of each possible decision, and assume projects choose the most profitable generator type for each resource class in each location. Like the capacity factor calculation, I assume generators take the best resource class. The ITC is a percent of this capital expenditure and does not need to be discounted or inflated to compare to the PTC.

Lastly, to calculate technology shares for each county, I use queue entry data from 2023 to 2025. For counties with queue entries, I set the technology shares of solar, wind, and batteries to the observed shares of each of those three technologies in the queue. For counties without queue entries, I assign the state share of each technology. For states with no queue data, I use the national share of each technology.

F.3 Formal Derivation of the Welfare Expression in Equation 6

This section presents the static model that derives Equation 6. The set up builds on Jacobsen et al. (2020), who base thier model on Harberger (1964).

Consider a representative consumer of county-by-technology electricity capacity x_j , measured in MW (e.g., MW of solar capacity in New Haven county), and a numeraire x_0 with price set to 1. The consumer maximizes its utility

$$\max_{x_0, x_1, \dots, x_J} U(x_1, \dots, x_J) + x_0 \quad \text{s.t.} \quad x_0 + \sum_j p_j x_j \leq M,$$

where there are J total county-technology combinations, and M is an exogenously specified income, and p_j is the price of x_j . Utility is assumed once differentiable, weakly increasing in all x_j .

A representative producer operates in a perfectly competitive environment to maximize profit

$$\max_{y_1, \dots, y_J} \sum_j (p_j + t_j) y_j - C(y_1, \dots, y_J),$$

where y_j is the supplied MW of county-technology j , C_j a total cost function, and t_j is a subsidy for j . The market clears when $x_j = y_j$ for all j .

Welfare is the sum of consumer surplus and producer surplus, minus the transfer required for the subsidies, plus an externality from consuming a unit of x_j , denoted ϕ_j :

$$\begin{aligned} W &= U(x_1, \dots, x_J) + M - \sum_j p_j x_j + \sum_j (p_j + t_j) x_j - C(x_1, \dots, x_J) + \sum_j \phi_j x_j - \sum_j t_j x_j \\ &= U(x_1, \dots, x_J) + M - C(x_1, \dots, x_J) + \sum_j \phi_j x_j. \end{aligned}$$

The first line subtitutes the consumer's budget constraint for x_0 and x_j for y_j according to the market clearing equation. The second line simplifies.

Toward proving Equation 6, first note that the change in welfare for a marginal change in the subsidy for a particular county-technology is

$$\frac{\partial W}{\partial t_j} = \sum_{k=1}^J \left(\frac{\partial U_k}{\partial x_k} - \frac{\partial C_k}{\partial x_k} + \phi_k \right) \frac{\partial x_k}{\partial t_j} = \sum_{k=1}^J (p_k - (p_k + t_k) + \phi_k) \frac{\partial x_k}{\partial t_j} = \sum_{k=1}^J (\phi_k - t_k) \frac{\partial x_k}{\partial t_j}.$$

The second equality follows from the consumer and producer's first order conditions in perfect competition.

Now, consider a change from some arbitrary tax vector $\tau = (\tau_1, \dots, \tau_J)$ to another arbitrary tax vector τ' . Let $\tilde{t}_j = (1 - \rho)\tau_j + \rho\tau'_j$ for all j . Differentiating the welfare function with respect to ρ and

integrating the result from $\rho = 0$ to $\rho = 1$ yields Equation 6.

$$\begin{aligned}
\frac{\partial W}{\partial \rho} &= \sum_j \sum_k \left(\frac{\partial U_j}{\partial x_j} - \frac{\partial C_j}{\partial x_j} + \phi_j \right) \frac{\partial x_j}{\partial t_k} \frac{\partial t_k}{\partial \rho} \\
&= \sum_j \sum_k (\phi_j - \tilde{t}_j) \frac{\partial x_j}{\partial t_k} \frac{\partial t_k}{\partial \rho} \\
&= \sum_j \sum_k ((\rho - 1)\tau_j - \rho\tau'_j + \phi_j) \frac{\partial x_j}{\partial t_k} (\tau'_k - \tau_k).
\end{aligned}$$

The second line follows from the first order conditions as before. The third substitutes in the definition and derivative of $\tilde{t}$. Finally, it is straightforward to integrate and rearrange this expression,

$$\int_0^1 \sum_j \sum_k ((\rho - 1)\tau_j - \rho\tau'_j + \phi_j) \frac{\partial x_j}{\partial t_k} (\tau'_k - \tau_k) d\rho = \frac{1}{2} \sum_j \sum_k (\phi_j - \tau_j + \phi_j - \tau'_j) \frac{\partial x_j}{\partial t_k} (\tau'_k - \tau_k).$$

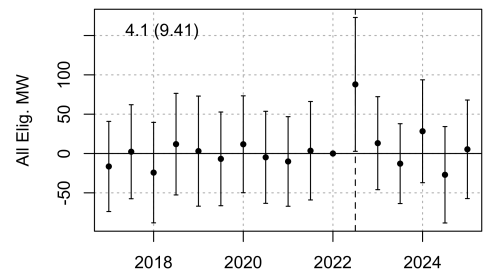
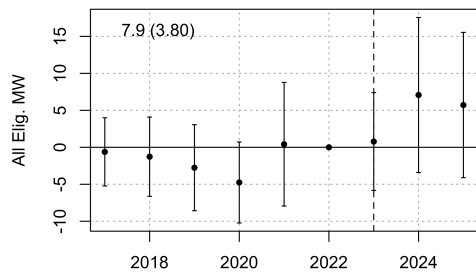
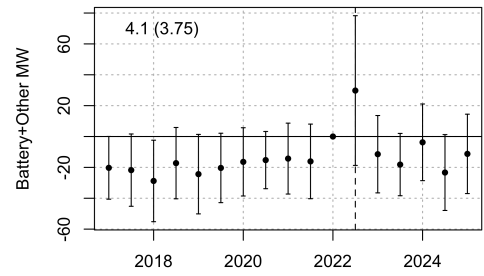
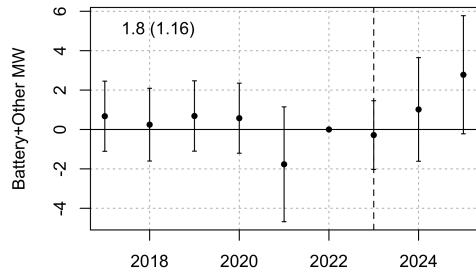
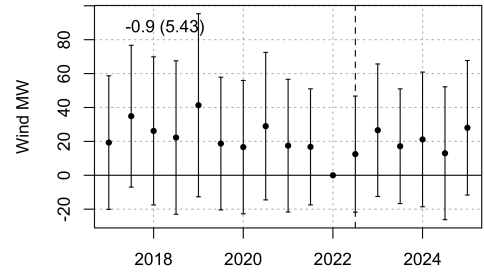
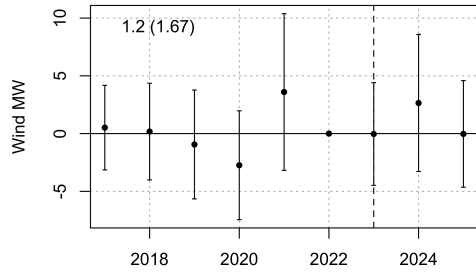
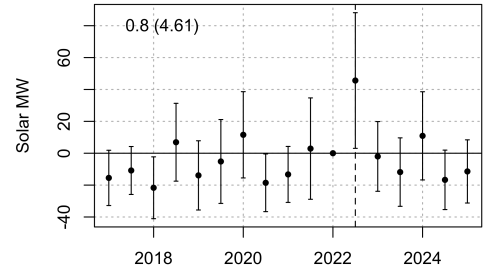
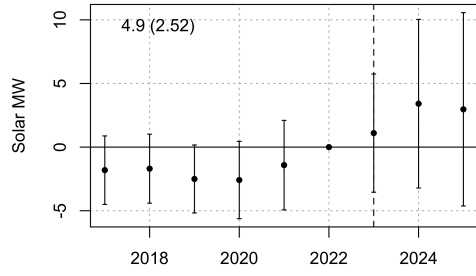
G Aggregate Trends in Capacity Additions and Queue Entry

Figure 24 shows trends in queue entry in the left column and capacity additions in the right column. The top row shows these statistics by technology type, and the bottom by Energy Community status. The graphs only show projects in areas that are Energy Communities and non-Energy Communities, dropping projects with uncertain Energy Community designation.²¹

Much fewer projects complete the queue than begin the queue; the y-axes of the queue entry and capacity addition charts differ by an order of magnitude. Queue entries grow quickly, reach a peak in 2022, and then contract after 2023, likely due to efforts by grid operators (such as the 2022 PJM two-year pause on queue entries) and the FERC 2023 mandate to reform the queue entry processes, both attempting to address the increasing backlog of queue entries. A growing share of capacity in queues is energy storage. This category includes both standalone and hybrid storage projects.

The dotted, dashed, and solid black lines show when the Energy Community Bonus was announced (July 2022), enacted (January 2023), and the guidance that detailed the locations of Energy Communities (April 2023). Energy Communities show a smaller reduction in queue entry after 2023 than do non-Energy Communities. There are more capacity additions in Energy Communities than in non-Energy Communities after 2023, but there is also evidence of pre-trends, motivating a careful empirical analysis.

²¹The Energy Community category includes projects in counties that are (1) completely covered by eligible coal tracts, (2) qualify as a fossil fuel employment Energy Community in all years 2023 to 2025, or (3) contain more than 10 brownfields. The non-Energy Community category are projects in counties with no coal tracts, never meeting the fossil employment definition, and with fewer than 10 brownfields.



(a) Capacity additions

(b) Queue entry

Figure 17: Spillover effect event studies and estimates with Energy Community neighbors as the treated group. Pooled treatment effect at the top-left with standard errors clustered at the county level.

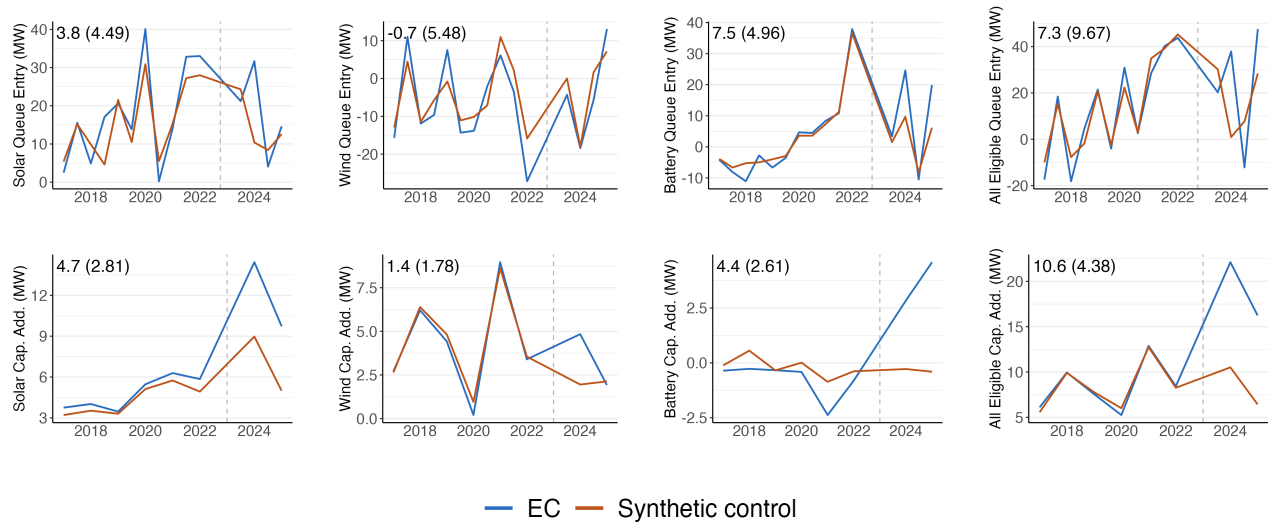


Figure 18: SDID results for queue entry (top row) and capacity addition (bottom row). Analyses restrict set the “treated” group to counties that neighbor Energy Communities from in the donor pool of controls of the primary specification. Energy Communities are dropped. The donor pool of controls for this specification are counties that are separated from Energy Communities by two or more other counties (i.e., third ring of counties around the Energy Communities and farther).

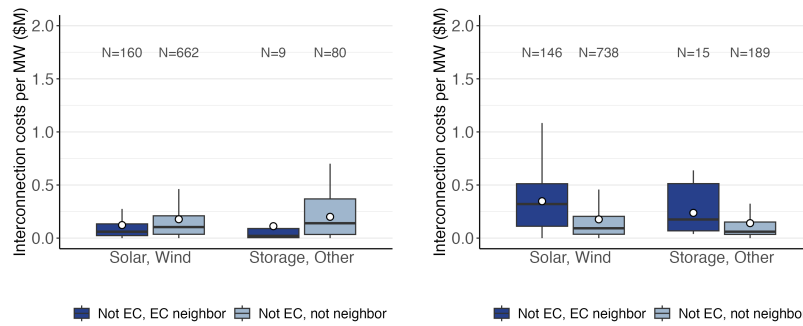


Figure 19: Interconnection costs per MW by technology before and after the Energy Community Bonus was enacted in 2023. The dark blue are neighbors of Energy Communities that are non-Energy Communities, meeting all of the restrictions that qualify a control county. The light blue are non-neighbor counties.

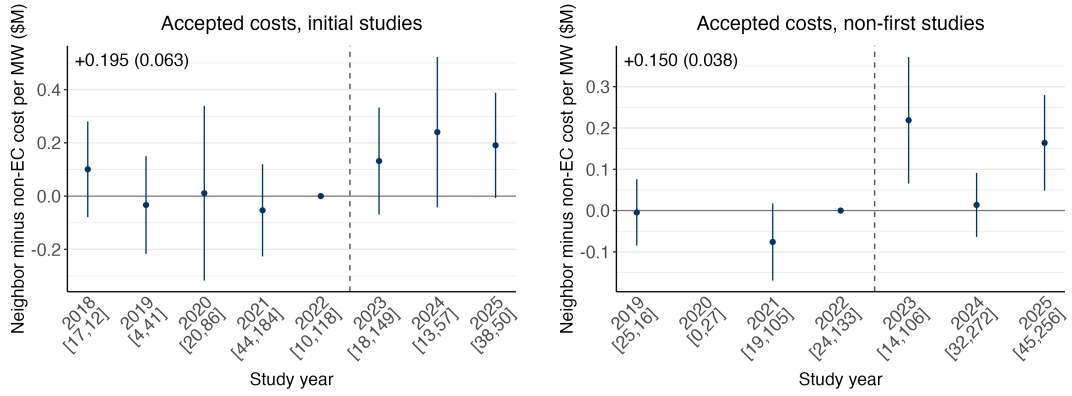
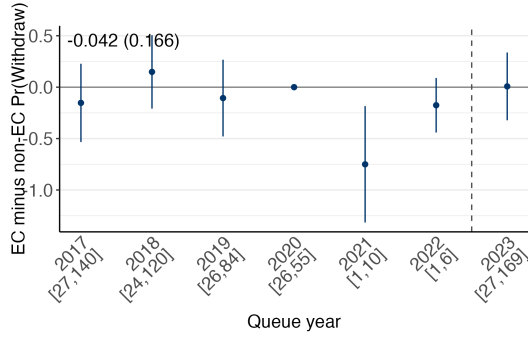
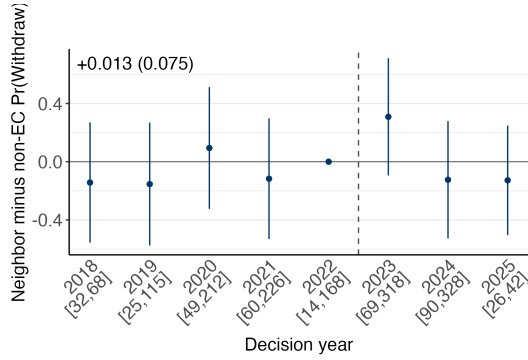


Figure 20: Event studies for data underlying Figure 19. Event studies include study year and technology fixed effects. Capacity-weighted. Numbers in brackets are number of project observations in Energy Communities and non-Energy Communities, respectively. Standard errors clustered at the county level.

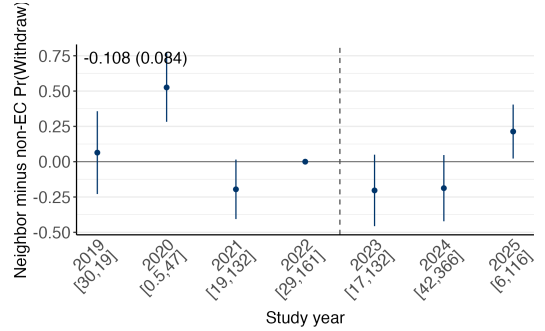


(a) Initial study, drop inframarginal



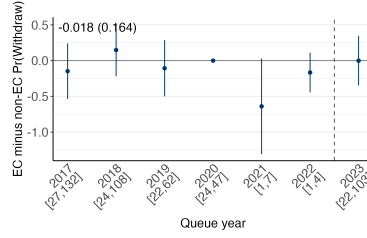
(c) Initial study, keep inframarginal

(b) Non-first studies, drop inframarginal (NA)

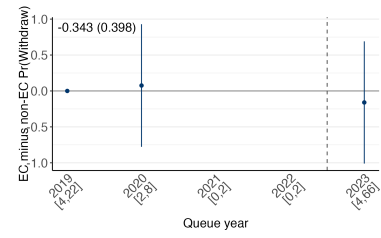


(d) Non-first studies, keep inframarginal

Figure 21: Event studies of probability of withdrawal after interconnection study for neighbors of Energy Communities compared to non-neighbors. NA indicates insufficient post-period observations to estimate. Capacity-weighted. Numbers in brackets are number of project observations in Energy Communities and non-Energy Communities, respectively.

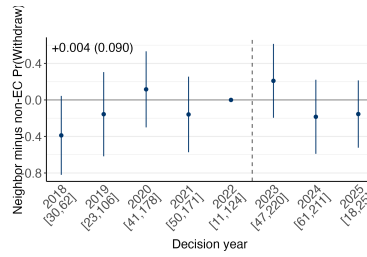


(a) Solar, wind

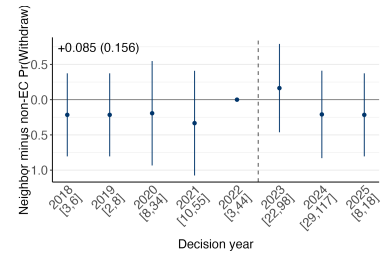


(b) Battery and other

Drop inframarginal, initial studies



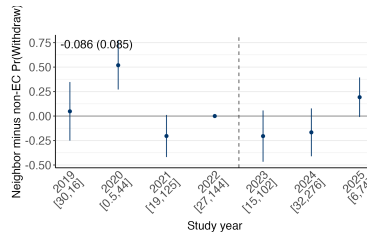
(c) Solar, wind



(d) Battery and other

Keep inframarginal, initial studies

Drop inframarginal, non-first studies



(g) Solar, wind

(f) Battery and other (NA)

Keep inframarginal, non-first studies

(h) Battery and other (NA)

Figure 22: NA indicates there are 2 or fewer observations in the post-period for the treated neighbors. Capacity-weighted. Numbers in brackets are number of project observations in Energy Communities and non-Energy Communities, respectively.

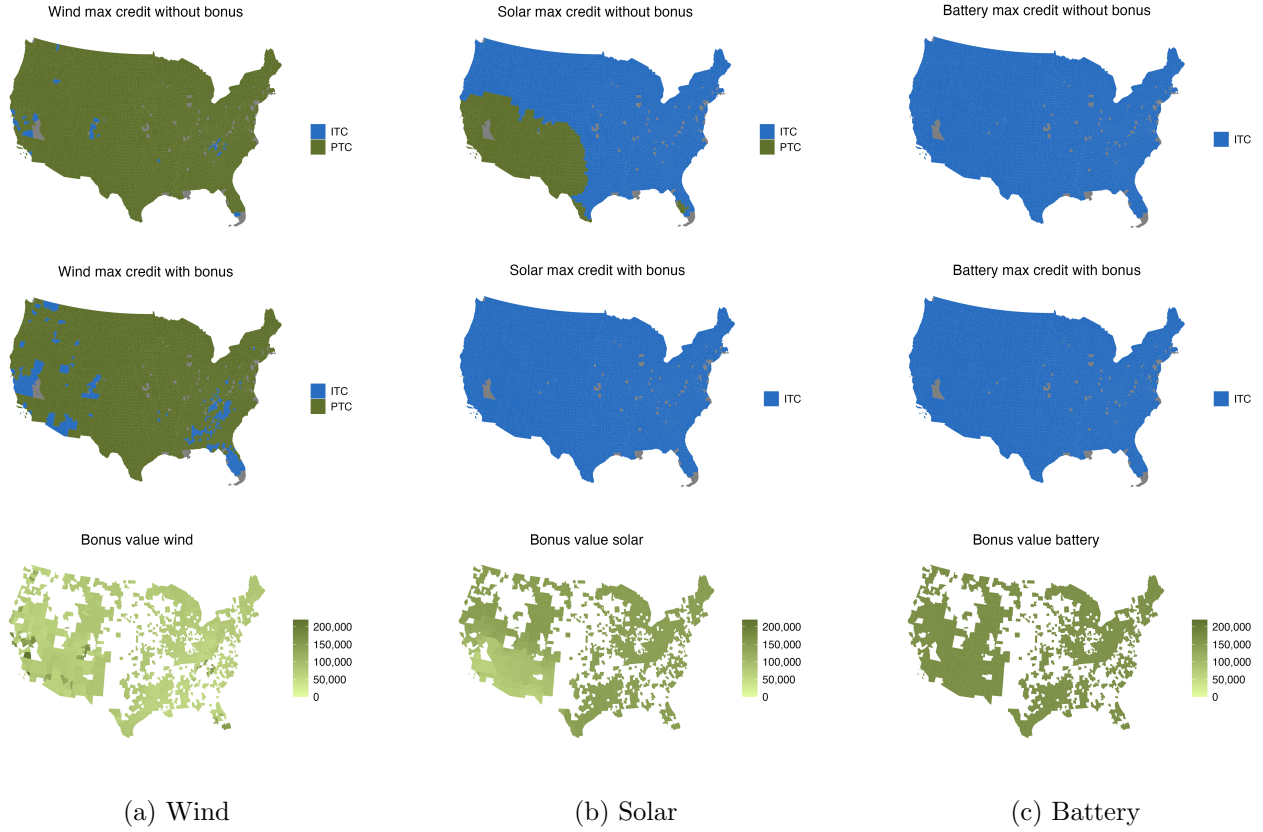


Figure 23: Tax credit choices for the baseline ITC and PTC (top row), with the Bonus (middle row), and the Energy Community Bonus value (bottom row). NA counties are those considered unsuitable for renewable electricity development in the capacity factor data, for example for being protected lands.

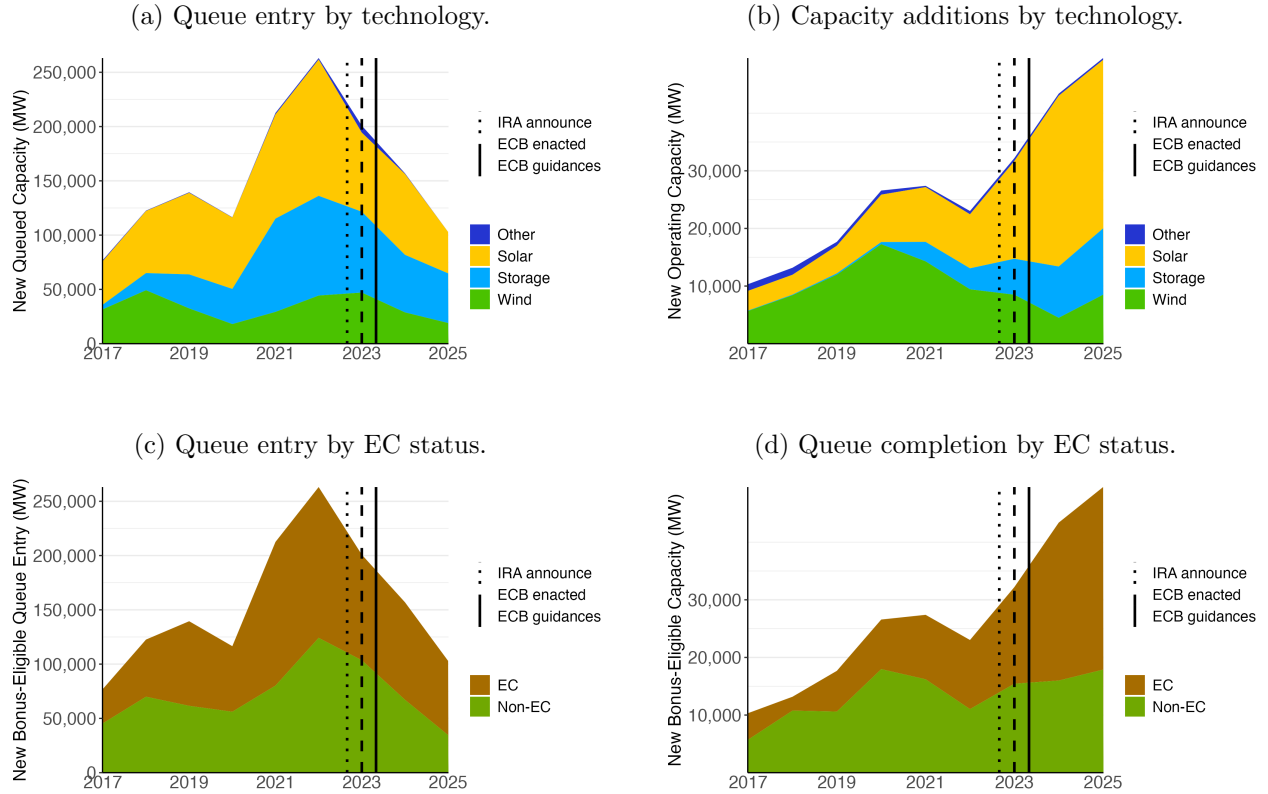


Figure 24: Panels (a) and (b) chart megawatts (MW) of queue entry and capacity additions over time, respectively, separated by technology. Both graphs only count MW of technologies eligible for the Inflation Reduction Act subsidies. The queue entry graph includes projects in the 1,666 counties that are completely covered by Energy Communities (EC) (38 percent of the sample) or with no Energy Community (non-EC) designation (62 percent of the sample). The capacity addition graph applies the same filter, yielding 2,000 counties, with the discrepancy due to unavailable queue data (37 percent EC, 63 percent non-EC). Panels (c) and (d) chart the same MW, separated by EC Bonus eligibility.